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Ravi et al.

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(54) **SEMICONDUCTOR EXFOLIATION METHOD**

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(51) **Int. Cl.**

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C30B 25/20	(2006.01)
C30B 29/36	(2006.01)
C30B 33/02	(2006.01)
H01L 21/02	(2006.01)
H10D 8/01	(2025.01)
H10D 8/60	(2025.01)
H10D 62/832	(2025.01)

(52) **U.S. Cl.**
CPC ***H01L 21/7813*** (2013.01); ***C30B 25/04*** (2013.01); ***C30B 25/20*** (2013.01); ***C30B 29/36*** (2013.01); ***C30B 33/02*** (2013.01); ***H01L 21/02378*** (2013.01); ***H01L 21/0243*** (2013.01); ***H01L 21/02529*** (2013.01); ***H10D 8/051*** (2025.01); ***H10D 8/60*** (2025.01); ***H10D 62/8325*** (2025.01)

(58) **Field of Classification Search**
CPC H01L 21/7813; H01L 21/02378; H01L 21/0243; H01L 21/02529; H01L 29/1608; H01L 29/6606; H01L 29/872; H01L 21/31144; H01L 21/02433; H01L 21/02444; H01L 21/02447; H01L 21/02494; H01L 21/02502; H01L 21/02642; H01L 21/02647; H01L 21/0262; C30B 25/04; C30B 25/20; C30B 29/36; C30B 33/02; C30B 33/06
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,770,507	B2	8/2004	Abe et al.
7,238,941	B2	7/2007	Tai et al.

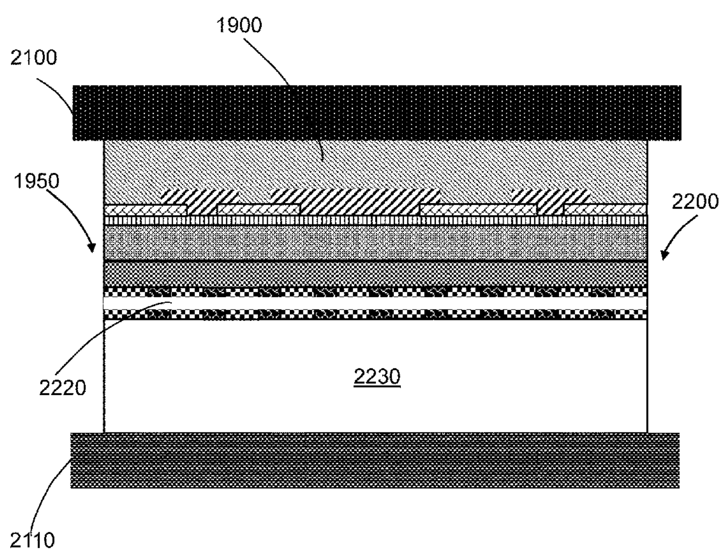
(Continued)

Primary Examiner — George R Koch
Assistant Examiner — Christopher C Caillouet

(57) **ABSTRACT**

A semiconductor substrate comprising a first epitaxial silicon carbide layer and a second silicon carbide epitaxial layer. At least one semiconductor device is formed in or on the second silicon carbide epitaxial layer. The semiconductor substrate is formed overlying a silicon carbide substrate having a surface comprising silicon carbide and carbon. An exfoliation process is used to remove the semiconductor substrate from the silicon carbide substrate. The carbon on the surface of the silicon carbide substrate supports separation. A portion of the silicon carbide substrate on the semiconductor substrate is removed after the exfoliation process. The surface of the silicon carbide substrate is prepared for reuse in subsequent formation of semiconductor substrates.

20 Claims, 30 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

7,790,226	B2	9/2010	Tai et al.
9,878,397	B2	1/2018	Hirata et al.
10,084,038	B2	9/2018	Vecino Vazquez et al.
10,304,739	B2	5/2019	Nishibayashi et al.
10,562,130	B1	2/2020	Donofrio et al.
10,576,585	B1	3/2020	Donofrio et al.
10,612,157	B2	4/2020	Kubota et al.
10,755,946	B2	8/2020	Hirata et al.
10,784,339	B2	9/2020	Vecino Vazquez et al.
10,825,733	B2	11/2020	Bhalla et al.
10,828,726	B2	11/2020	Hirata et al.
10,870,176	B2	12/2020	Hirata
10,872,757	B2	12/2020	Hirata
10,872,758	B2	12/2020	Hirata
10,903,078	B2	1/2021	Schulze et al.
10,978,311	B2	4/2021	Drescher et al.
11,031,483	B2	6/2021	Schulze et al.
11,081,393	B2	8/2021	Beyer et al.
11,211,459	B2	12/2021	Brockmeier et al.
2003/0056718	A1	3/2003	Kawahara et al.
2021/0225659	A1	7/2021	Drescher et al.
2021/0225709	A1	7/2021	Rieske et al.
2023/0009693	A1*	1/2023	Gao H01L 21/7806

* cited by examiner

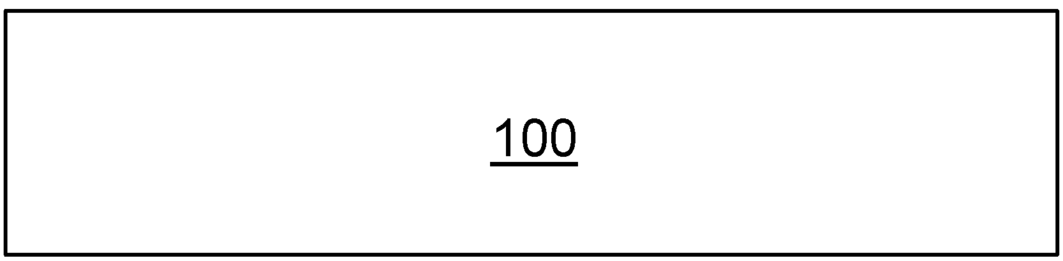


FIG. 1

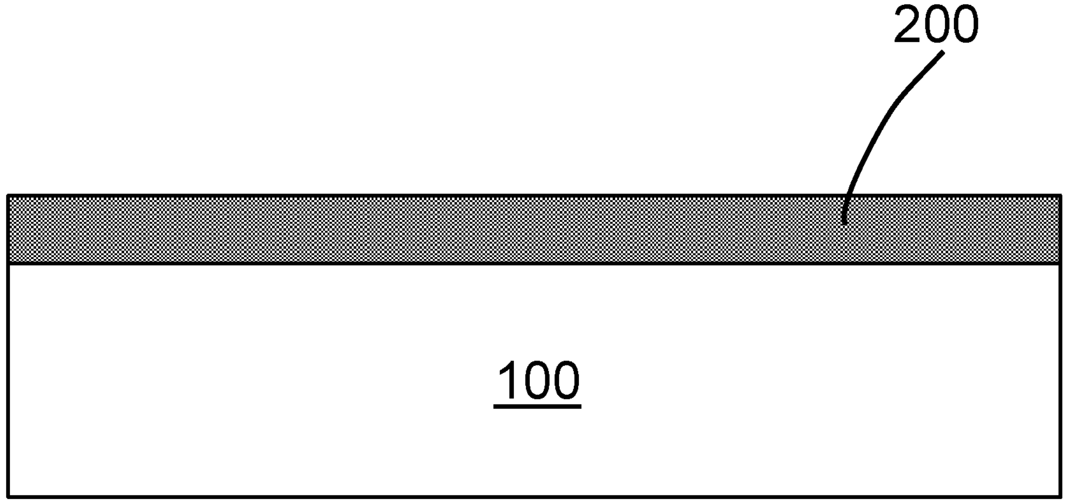


FIG. 2

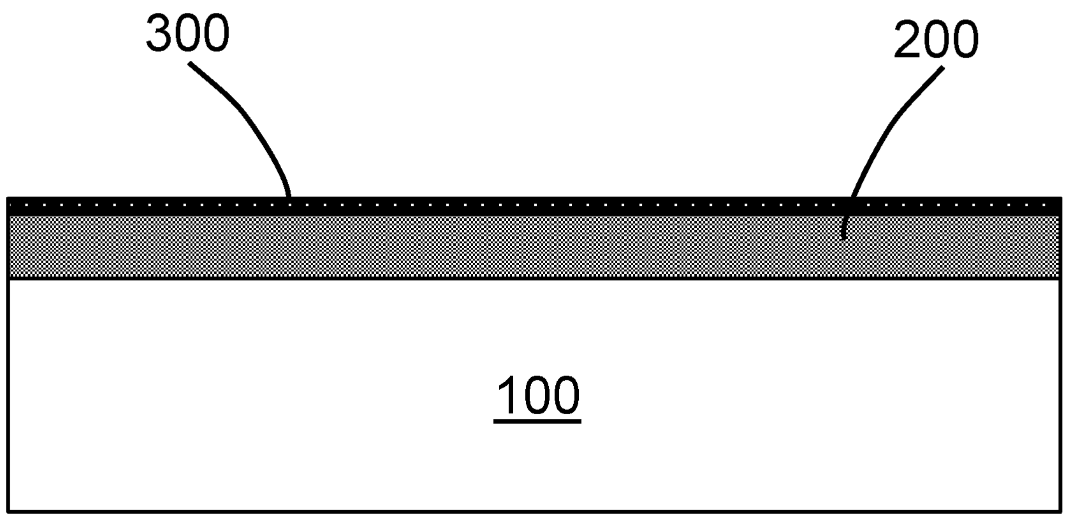


FIG. 3

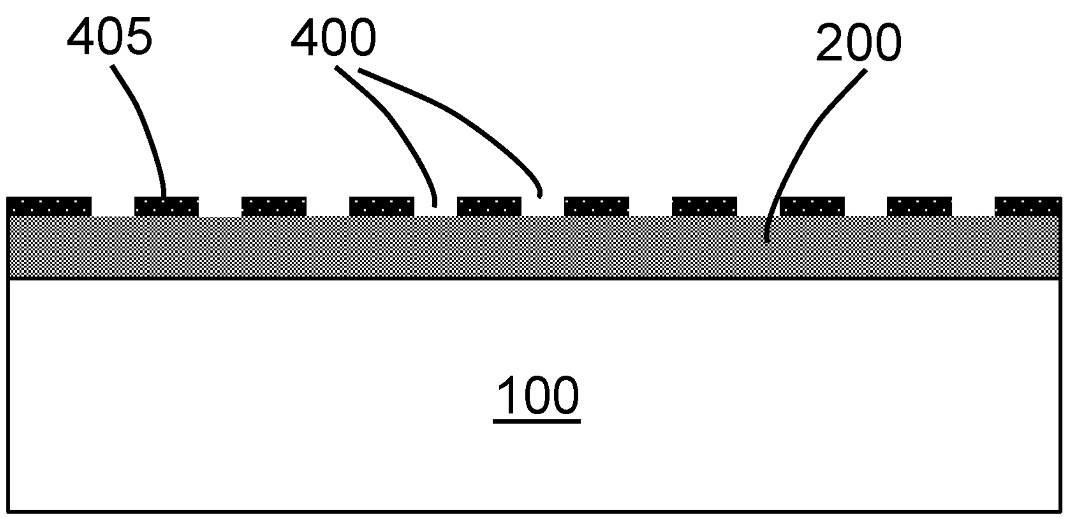


FIG. 4

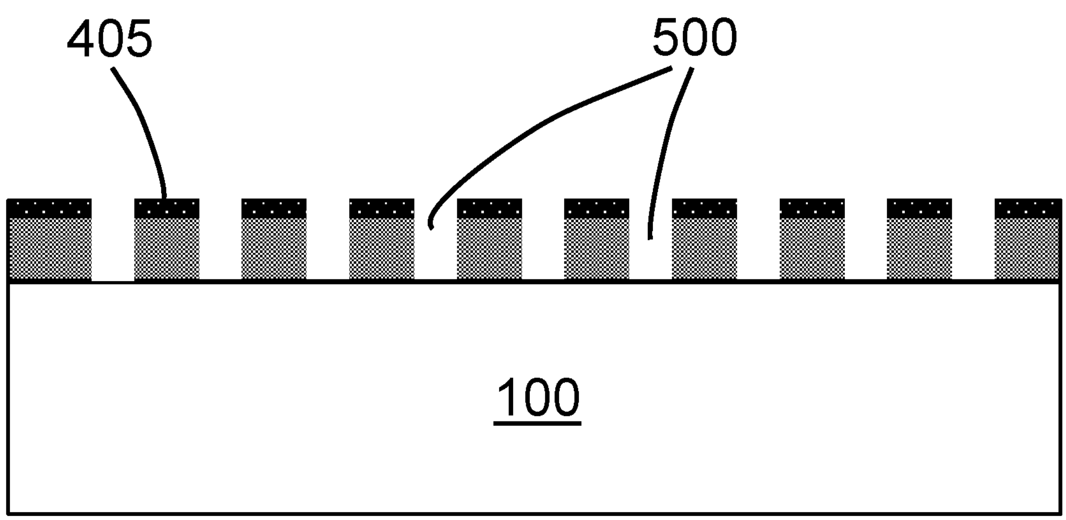


FIG. 5

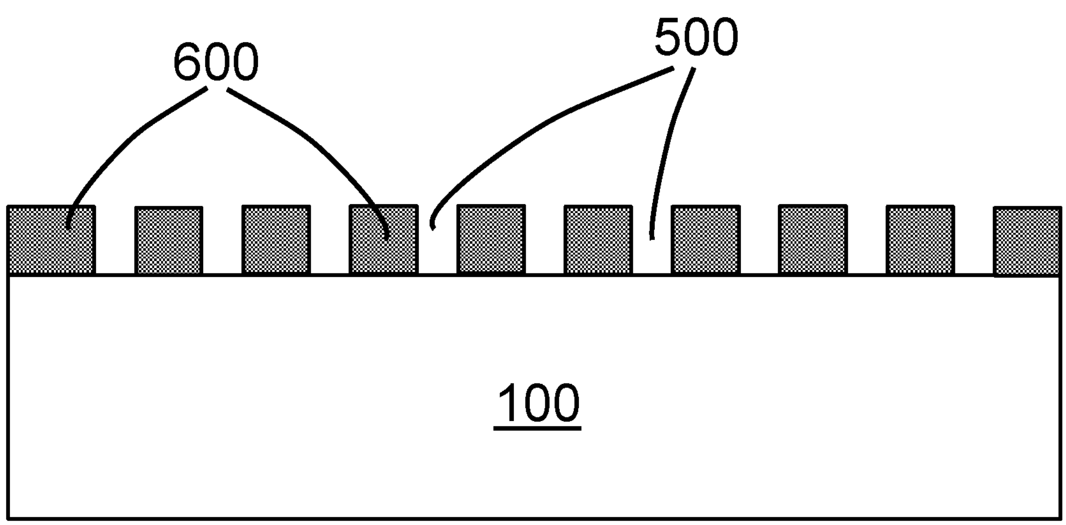


FIG. 6

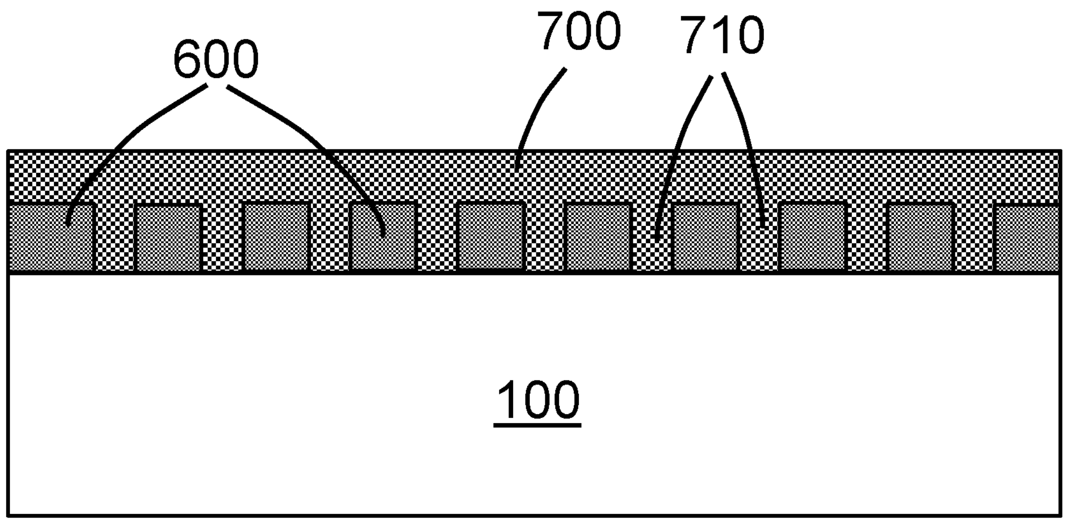


FIG. 7

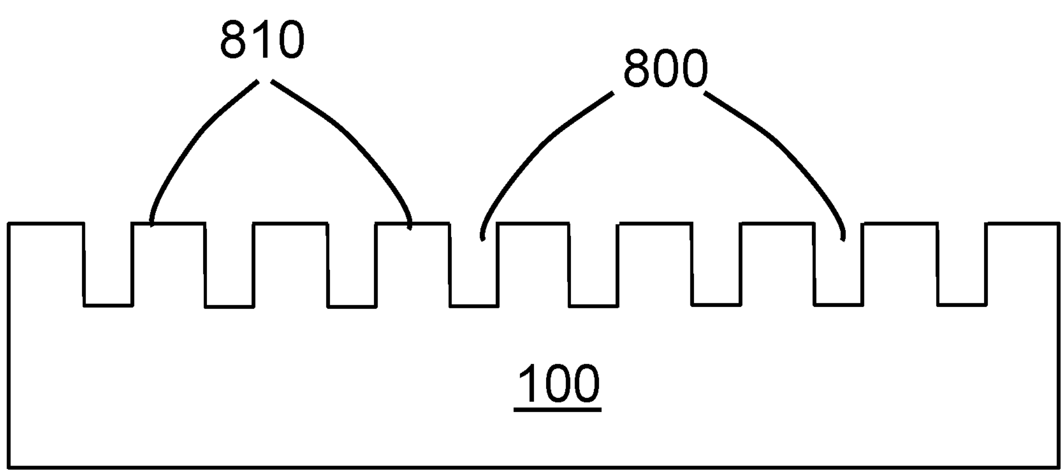


FIG. 8

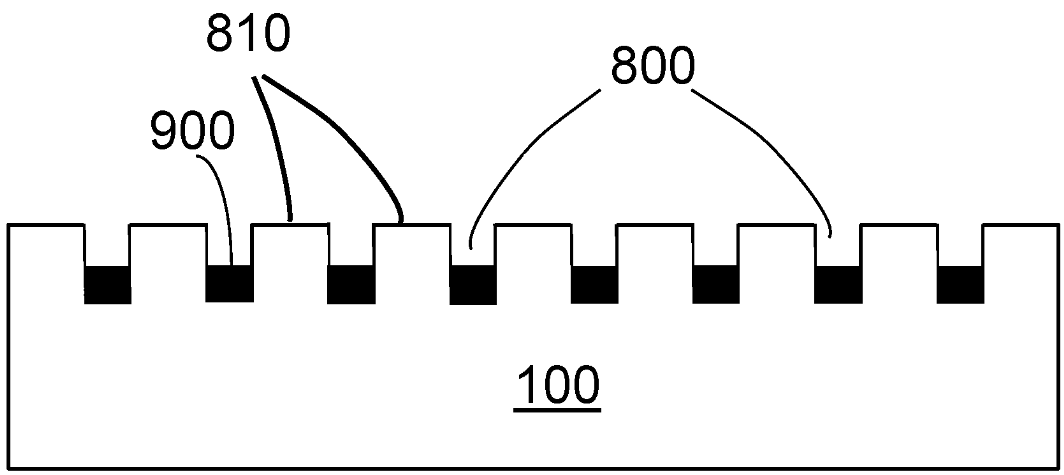


FIG. 9

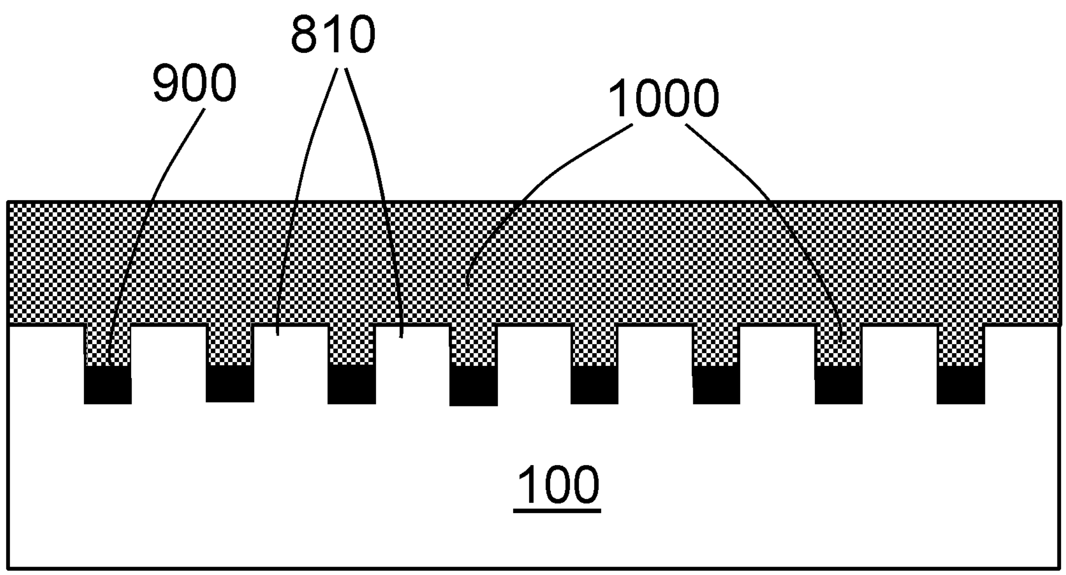


FIG. 10

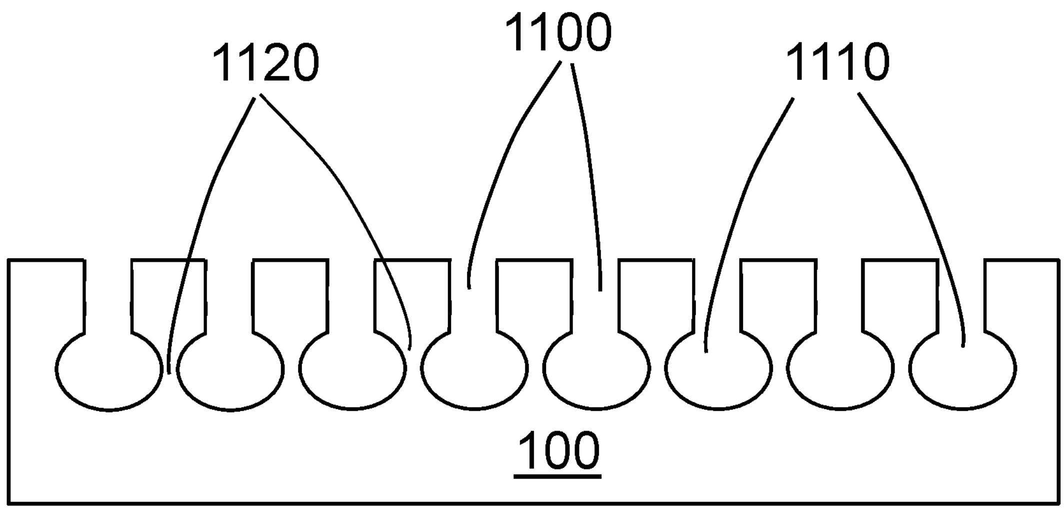


FIG. 11

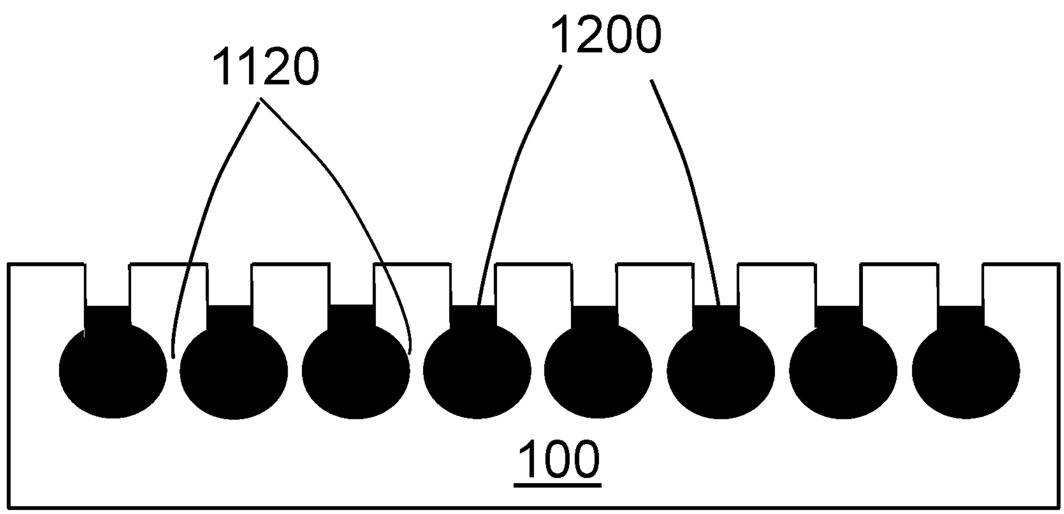


FIG. 12

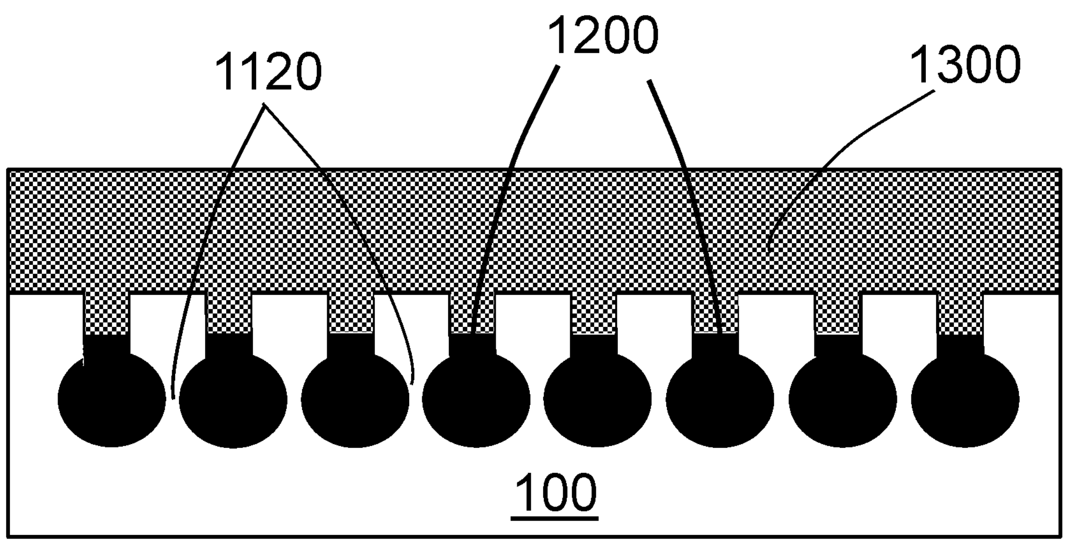


FIG. 13

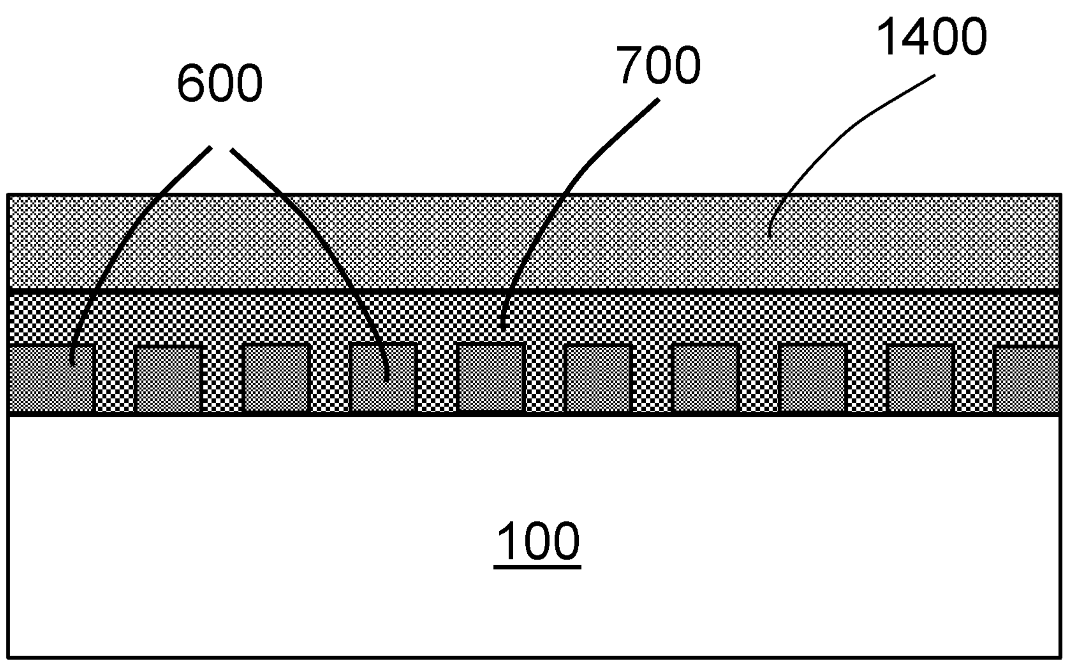


FIG. 14

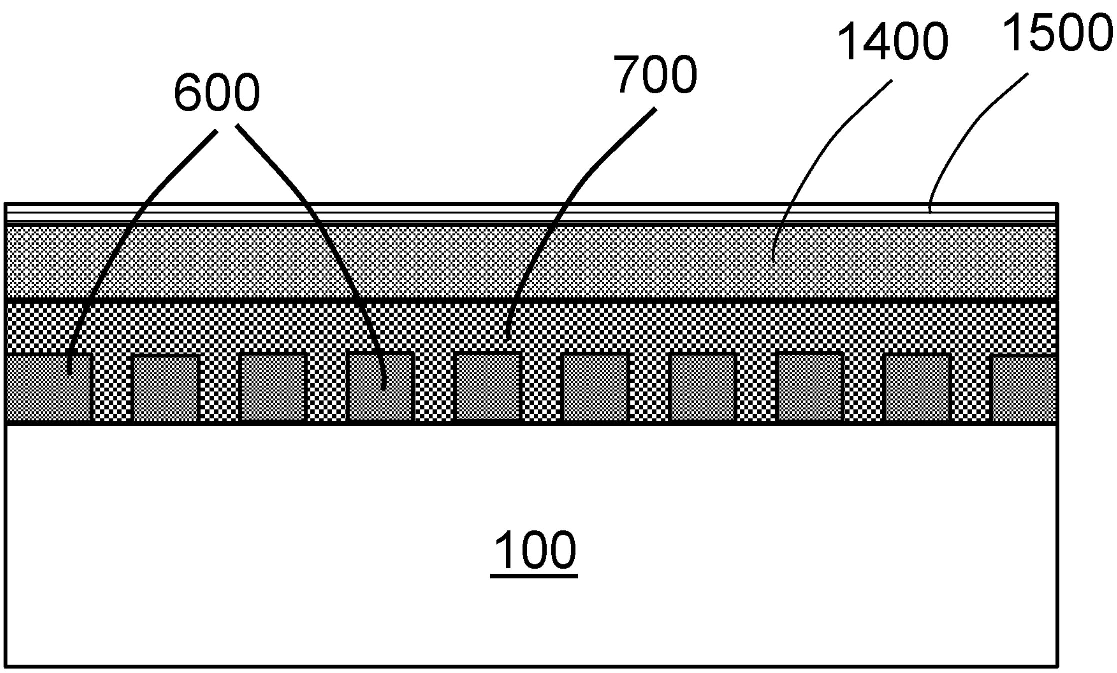


FIG. 15

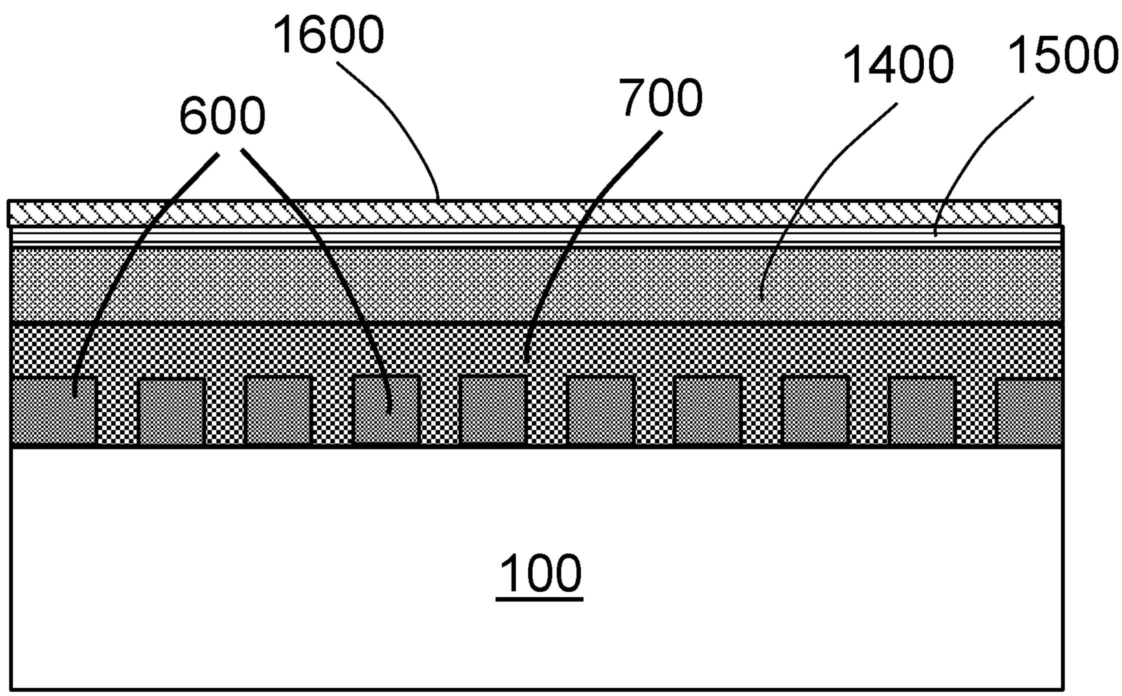


FIG. 16

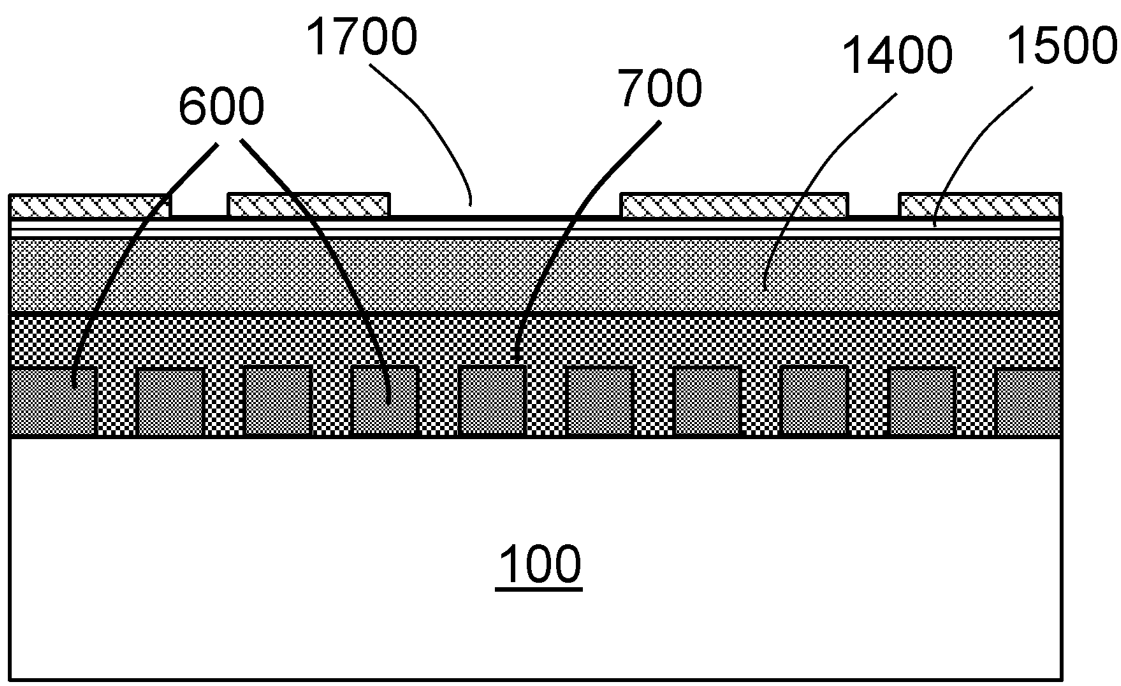


FIG. 17

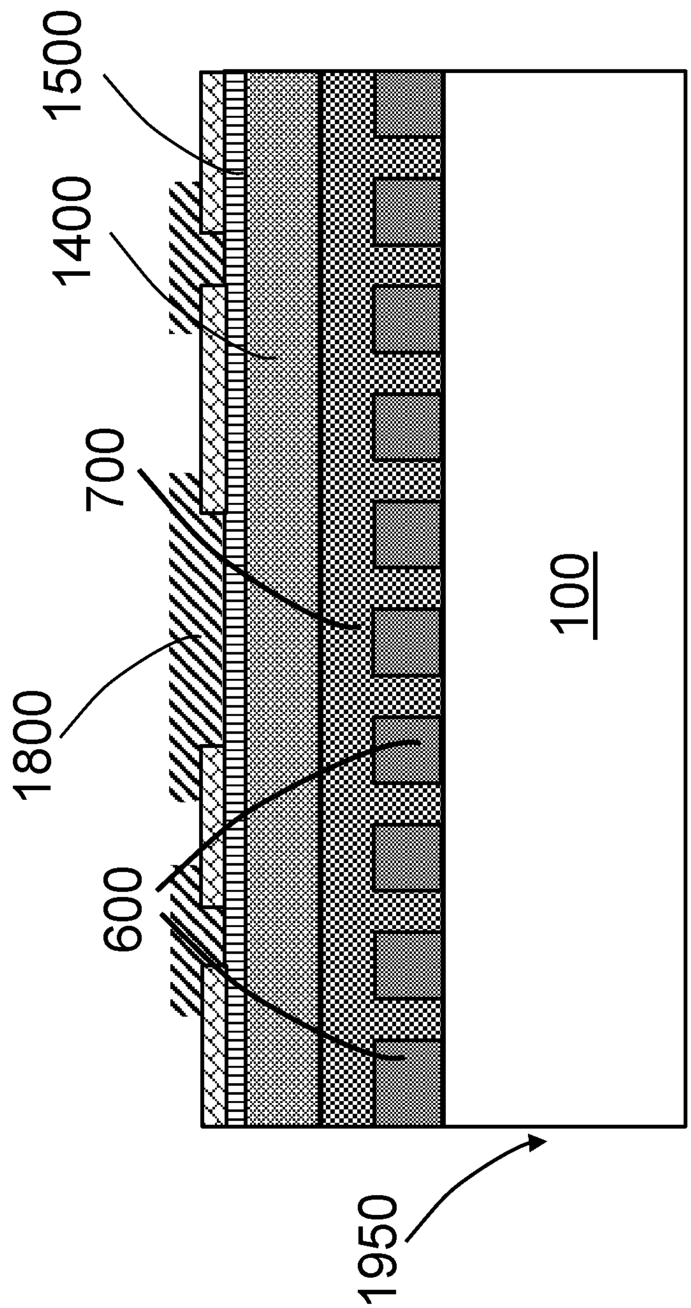


FIG. 18

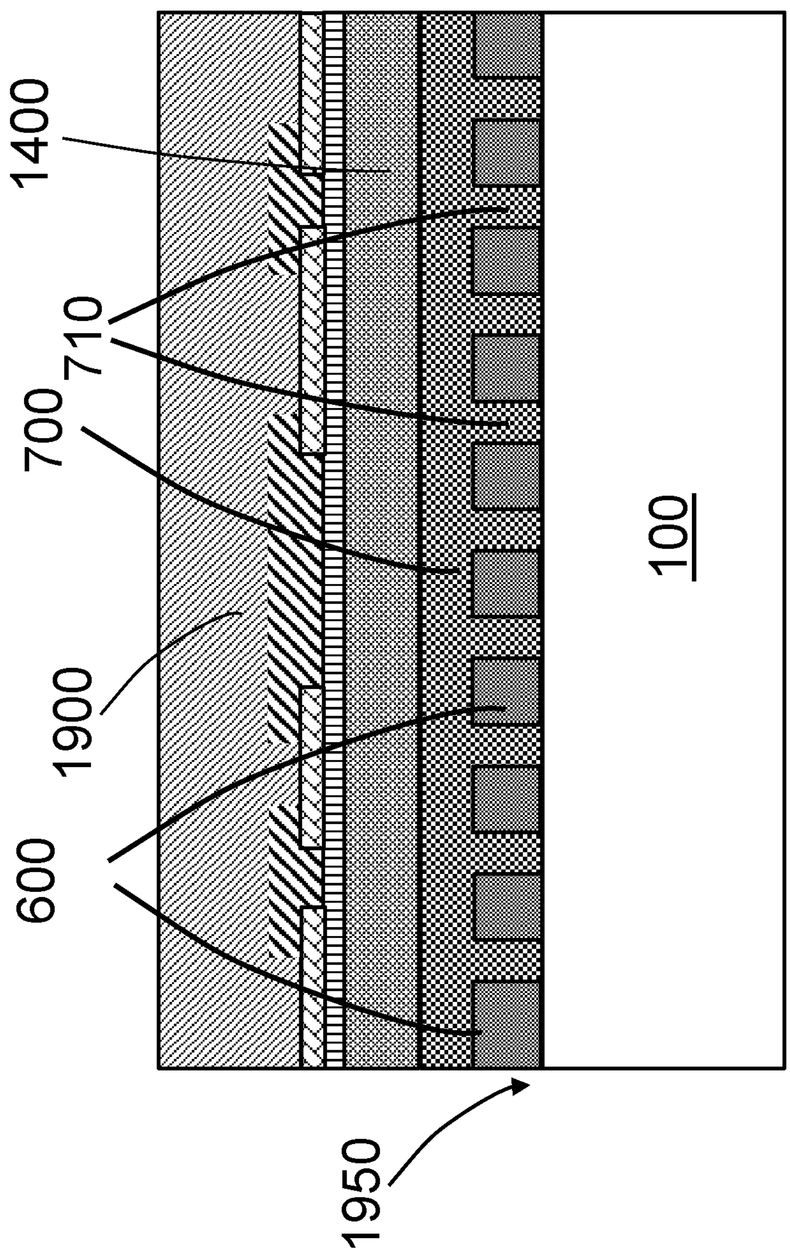


FIG. 19

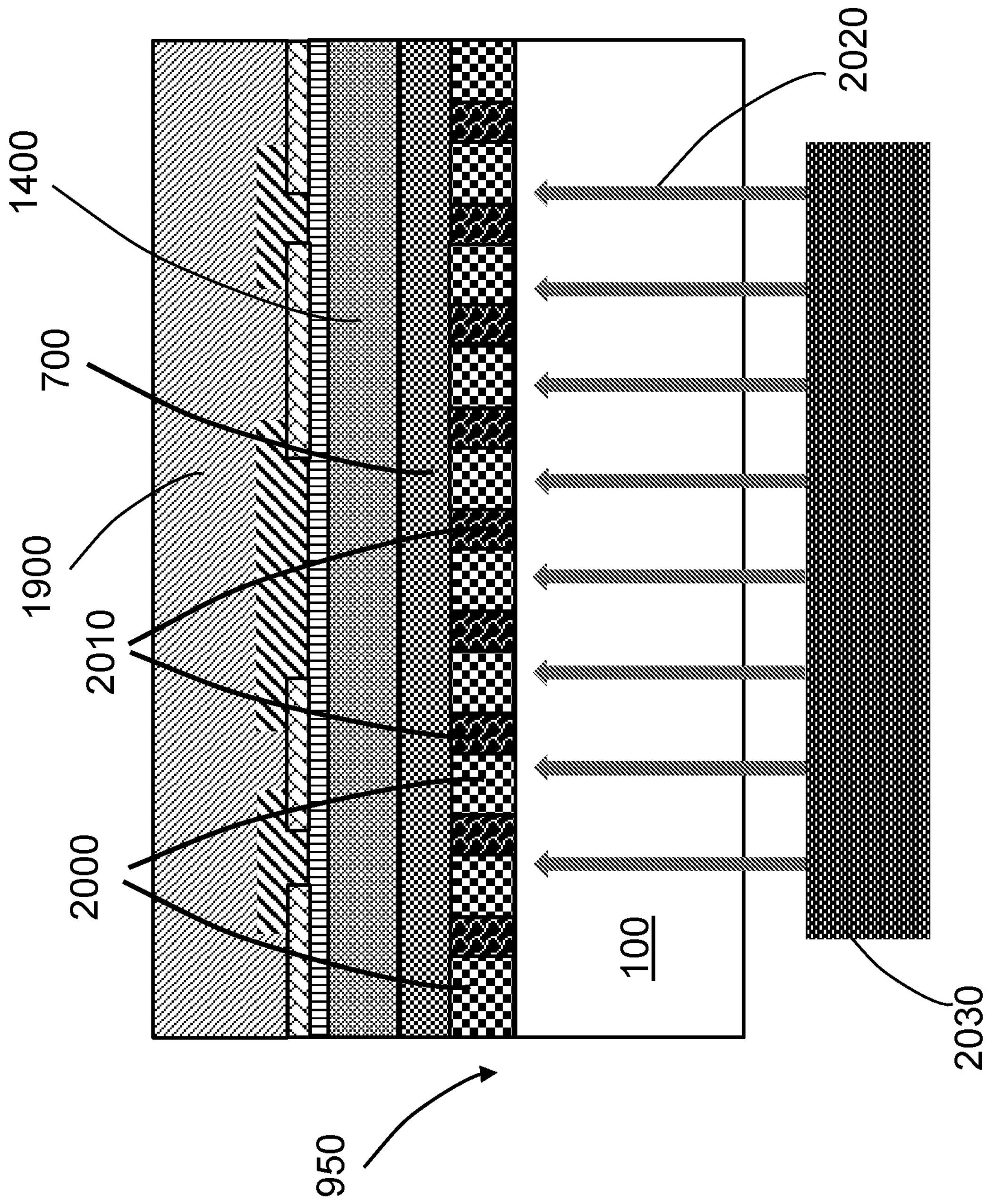


FIG. 20

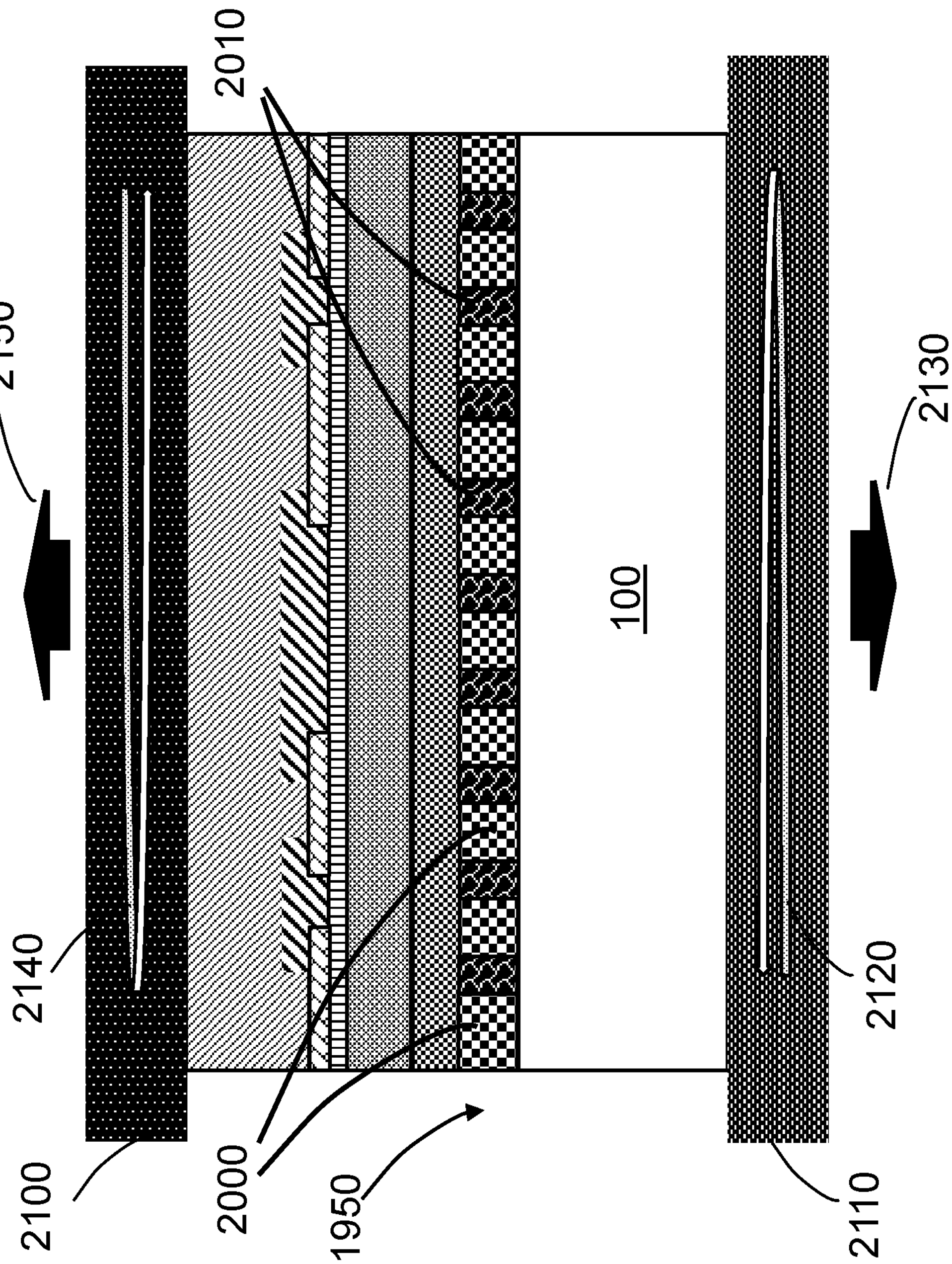


FIG. 21

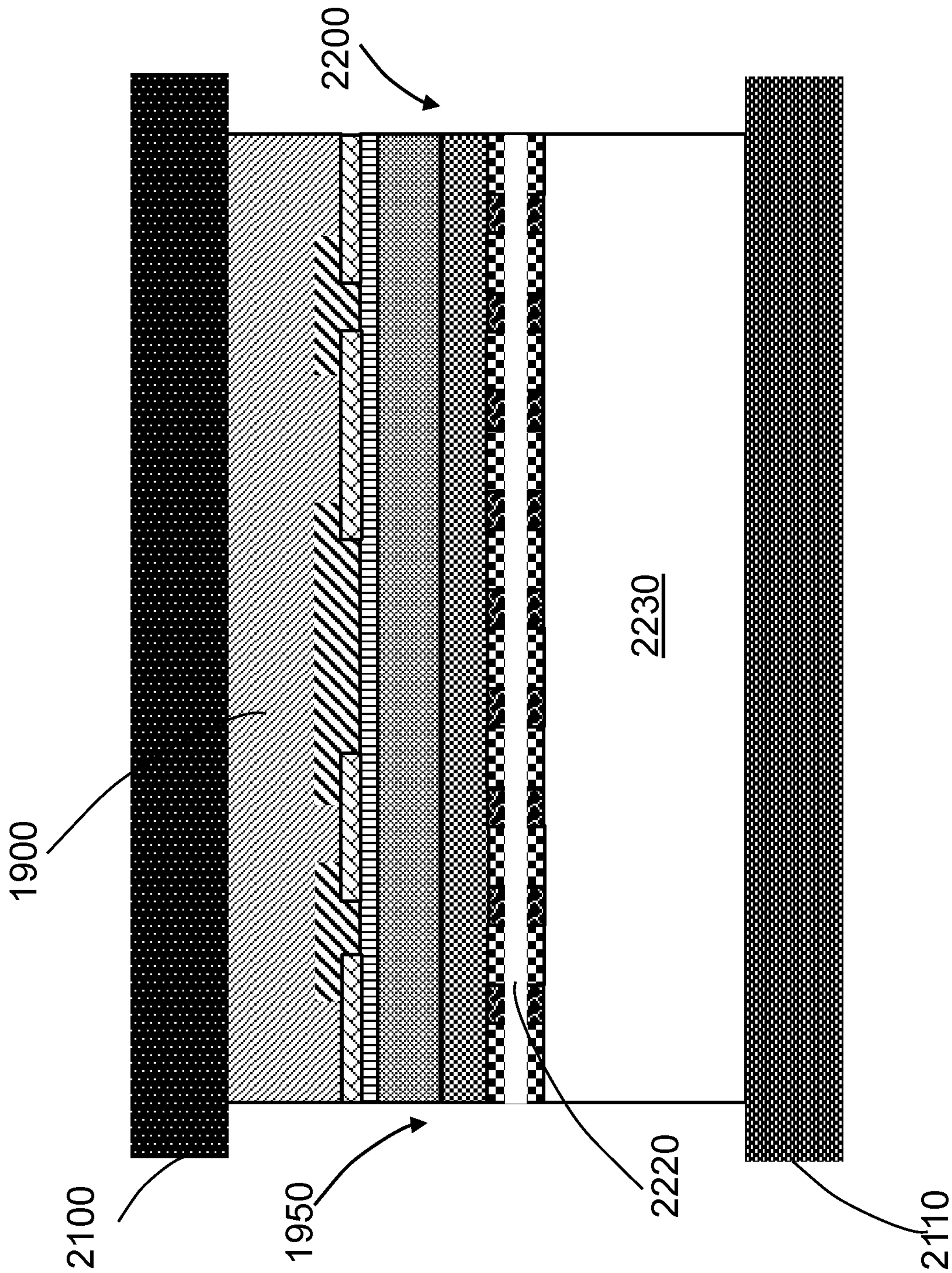


FIG. 22

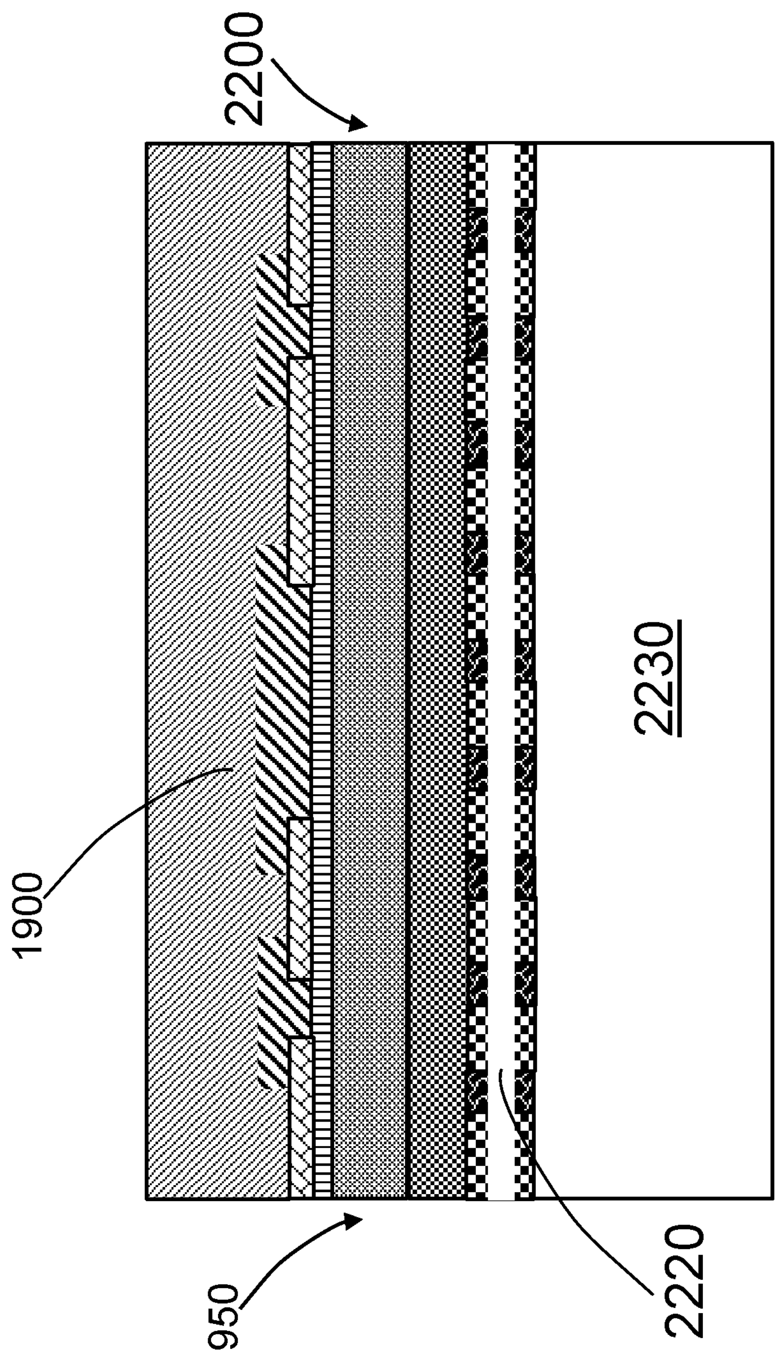


FIG. 23

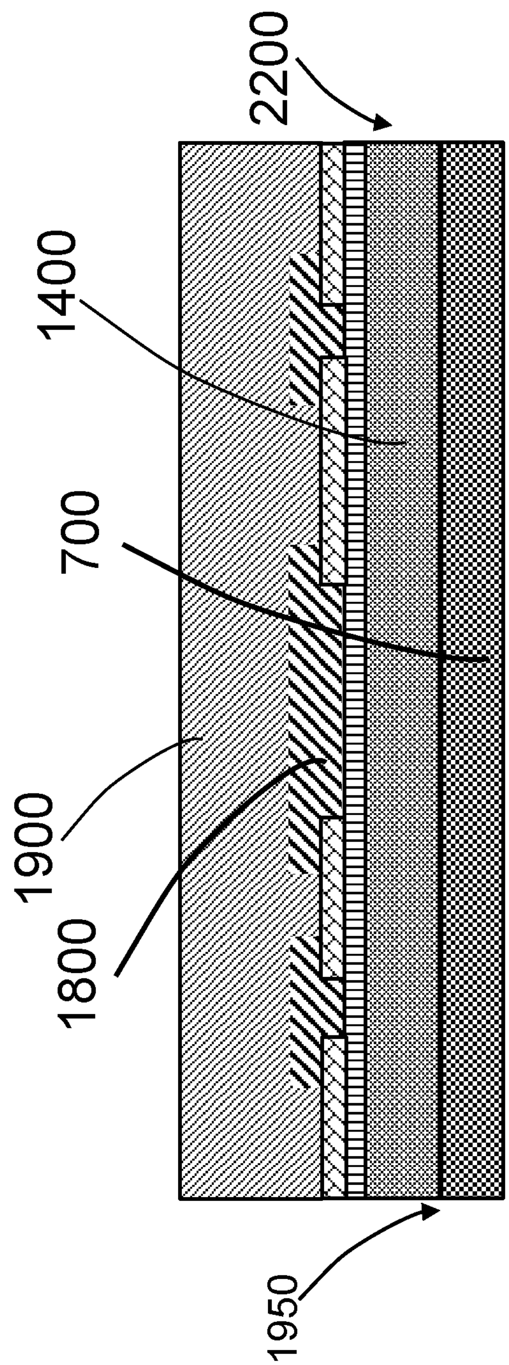


FIG. 24

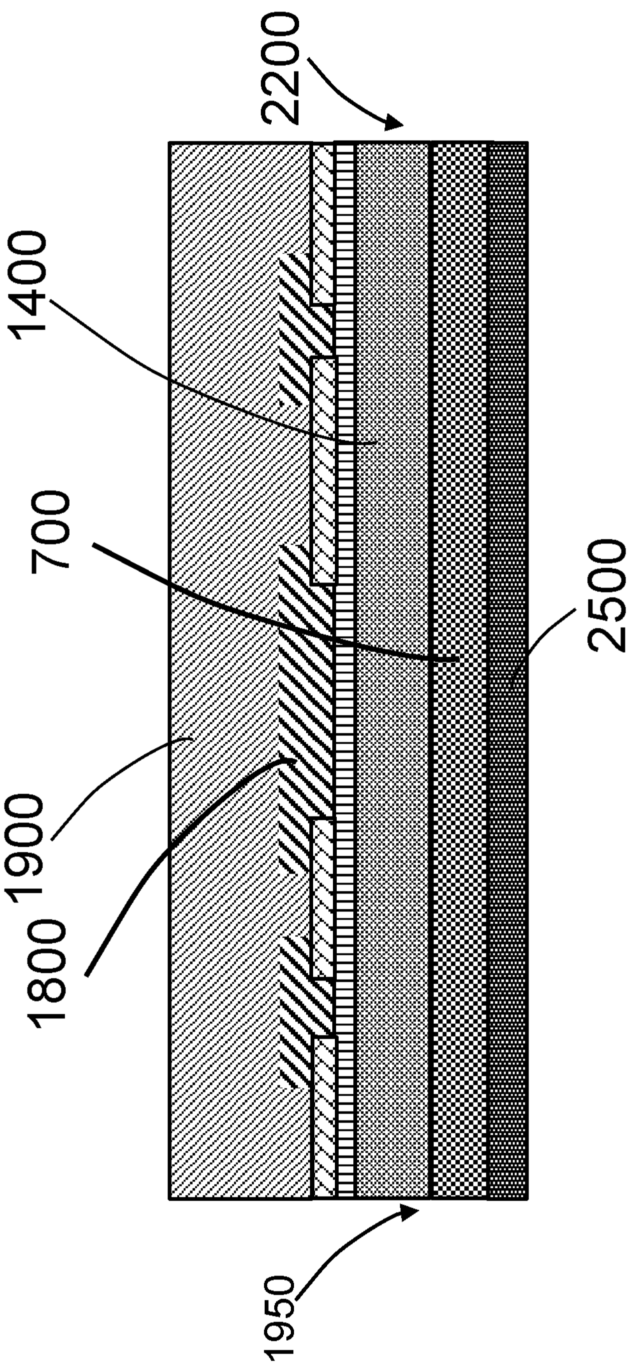


FIG. 25

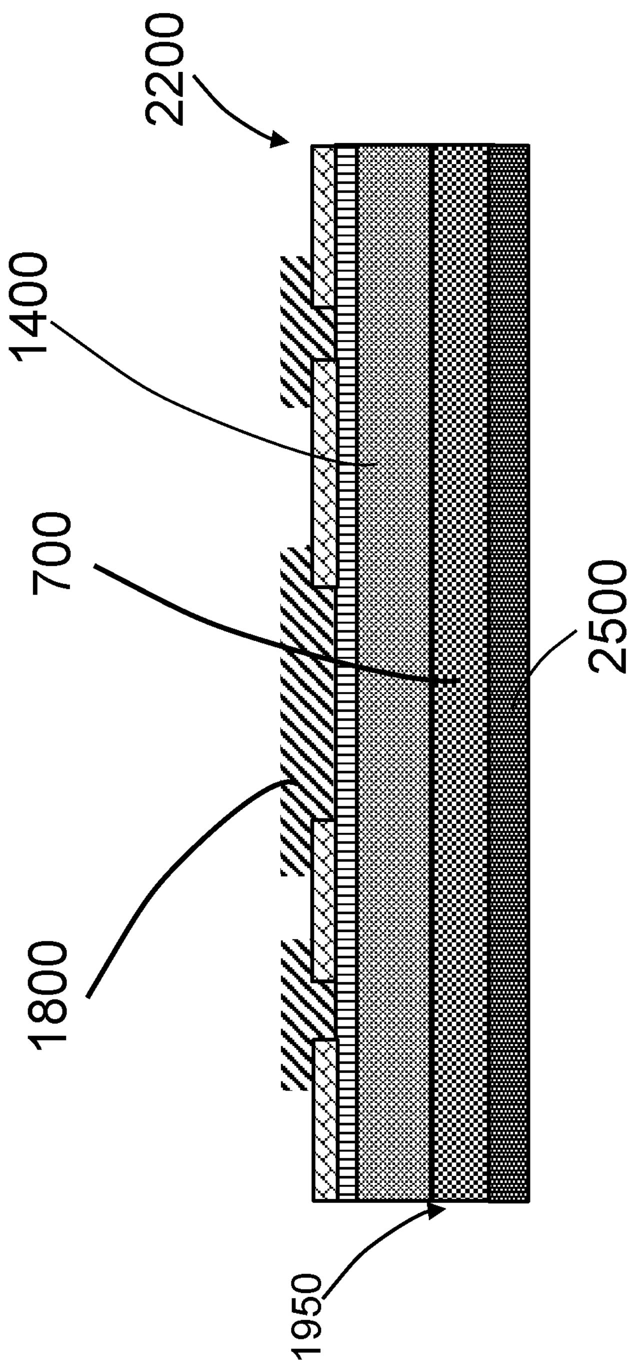


FIG. 26

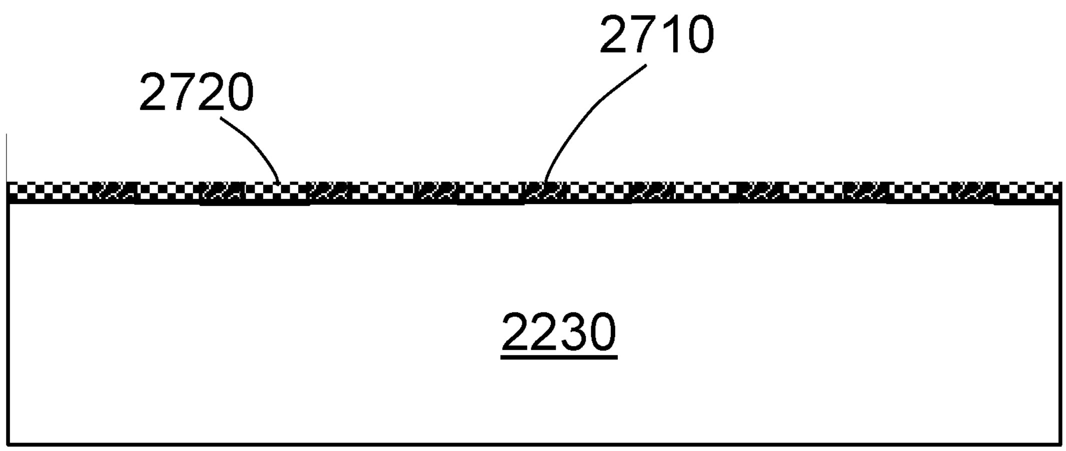


FIG. 27

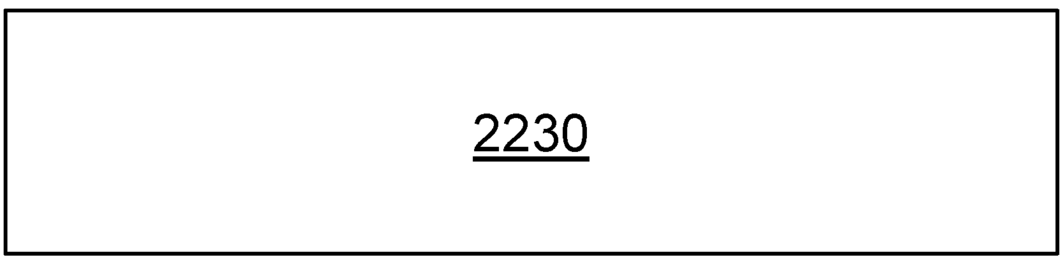


FIG. 28

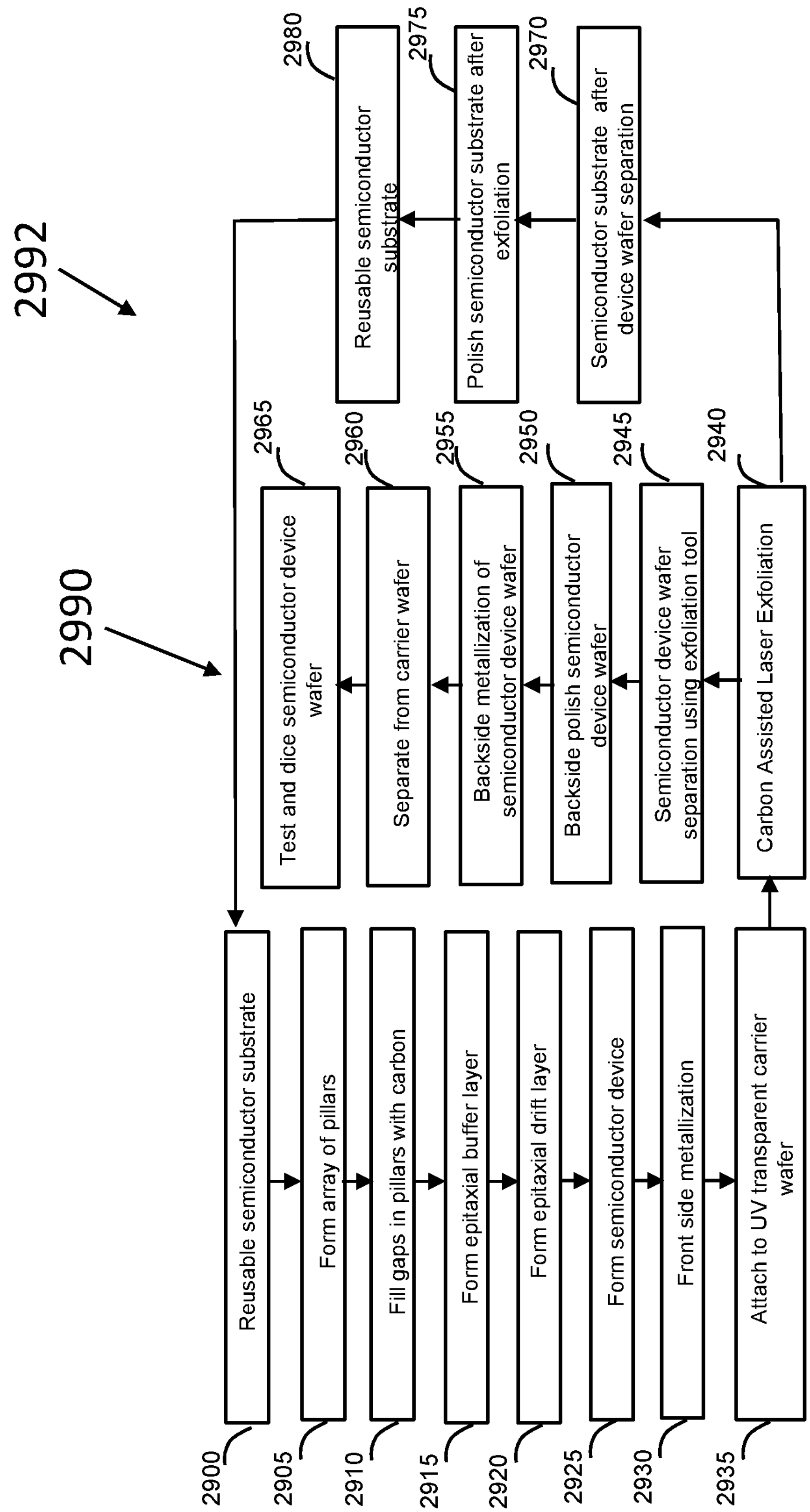

2992
2990
Reusable semiconductor substrate
Form array of pillars
Fill gaps in pillars with carbon
Form epitaxial buffer layer
Form epitaxial drift layer
Form semiconductor device
Front side metallization
Attach to UV transparent carrier wafer
Carbon Assisted Laser Exfoliation
Semiconductor device wafer separation using exfoliation tool
Backside polish semiconductor device wafer
Backside metallization of semiconductor device wafer
Separate from carrier wafer
Test and dice semiconductor device wafer
Semiconductor substrate after device wafer separation
Polish semiconductor substrate after exfoliation
Reusable semiconductor substrate
2900
2905
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FIG. 29

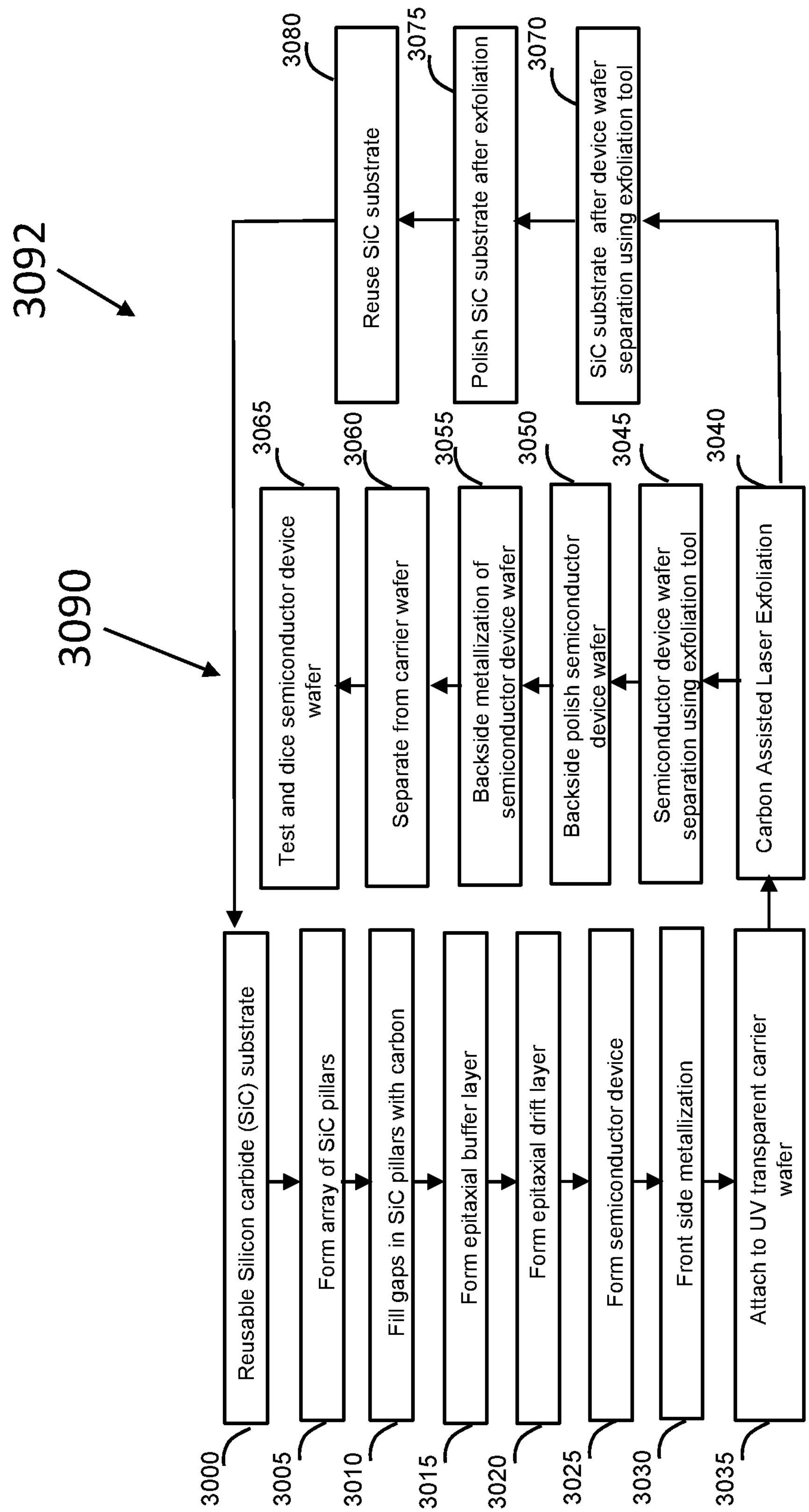
3090
3092
3000
Reusable Silicon carbide (SiC) substrate
3005
Form array of SiC pillars
3010
Fill gaps in SiC pillars with carbon
3015
Form epitaxial buffer layer
3020
Form epitaxial drift layer
3025
Form semiconductor device
3030
Front side metallization
3035
Attach to UV transparent carrier wafer
3040
Carbon Assisted Laser Exfoliation
3045
Semiconductor device wafer separation using exfoliation tool
3050
Backside polish semiconductor device wafer
3055
Backside metallization of semiconductor device wafer
3060
Separate from carrier wafer
3065
Test and dice semiconductor device wafer
3070
SiC substrate after device wafer separation using exfoliation tool
3075
Polish SiC substrate after exfoliation
3080
Reuse SiC substrate

FIG. 30

SEMICONDUCTOR EXFOLIATION METHOD

FIELD

This invention relates to semiconductor substrates and, in particular to forming a semiconductor substrate comprising one or more epitaxial layers.

BACKGROUND

The use of wide bandgap (WBG) semiconductors has increased dramatically in recent years in power electronics. Their ability to operate efficiently at higher voltages, powers, temperatures, and switching frequencies has enabled reduced cooling requirements, lower part counts, and the use of smaller passive components. WBG-based power electronics can further reduce the footprint and potentially the system cost of various renewable energy electrical equipment such as motor drivers and inverters.

Among the WBG semiconductors for power electronics, Silicon Carbide (SiC) is now increasingly used for high voltage drivers (>1200V) whereas Gallium Nitride (GaN) has been experiencing increased use in both higher power and higher frequency applications. However, unlike silicon, the cost of a final device for WBG semiconductor devices is dominated by the cost of the materials. The materials include the substrate and the active layer grown by Epitaxy. The substrate by itself contributes to over half of the cost of a finished WBG semiconductor device.

From the substrate standpoint, 4H-Silicon carbide (SiC) Single Crystal Substrates have been used for both SiC and GaN devices since SiC and GaN epitaxial layers can be grown with reduced defects on SiC substrates. The GaN substrate, on the other hand, is very expensive to grow defect free and has not kept up with scaling size increases afforded with SiC substrates. While the SiC substrate quality has dramatically improved in the recent years, the cost has not come down since substrate fabrication is a complex process starting with vapor phase ingot growth followed by ingot cropping, then wire sawing of individual wafers, and finally grinding and polishing of the substrate, and as of now, there has been no proven practical method to eliminate any of these foregoing steps. As a semiconductor substrate for WBG semiconductors is being produced and devices that use high currents are fabricated, defects play a larger role and are magnified because die sizes are larger and any defect will contribute to more significant yield loss and potential lower reliability. Therefore, to maximize die yield, any cost reduction activity regarding the substrate is paramount while also maintaining low defect densities in the active device epitaxial layer. The effect of material defects is magnified due to larger die sizes which would therefore incorporate more material defects which would in each contribute to more significant yield loss and potentially lower reliability. Therefore, to maximize die yield and cost reduction activity regarding the substrate while also maintaining low defect densities in the active epitaxial layers is paramount.

Accordingly, it is desirable to provide methods to manufacture WBG semiconductors that overcome the thin substrate limitation and reduce the contribution of the substrate to the final die with minimal effect to the yield or performance parameters of the final WBG semiconductor.

BRIEF DESCRIPTION OF THE DRAWINGS

Various features of the system are set forth with particularity in the appended claims. The embodiments herein, can

be understood by reference to the following description, taken in conjunction with the accompanying drawings, in which:

FIG. **1** illustrates a reusable silicon carbide (SiC) single crystal substrate in accordance with an example embodiment;

FIG. **2** illustrates a carbon layer over the silicon carbide substrate in accordance with an example embodiment;

FIG. **3** illustrates a hard mask overlying a carbon layer overlying a silicon carbide substrate in accordance with an example embodiment;

FIG. **4** illustrates a patterned hard mask overlying a carbon layer overlying a silicon carbide substrate in accordance with an example embodiment;

FIG. **5** illustrates a plurality of trenches etched in carbon layer forming patterned carbon regions overlying a silicon carbide substrate in accordance with an example embodiment;

FIG. **6** illustrates the silicon carbide substrate with plurality of trenches in carbon layer forming patterned carbon regions after removal of patterned hard mask in accordance with an example embodiment;

FIG. **7** illustrates an epitaxially grown layer with merged epitaxial lateral overgrowth (MELO) over patterned carbon regions in silicon carbide substrate in accordance with an example embodiment;

FIG. **8** illustrates a reusable silicon carbide (SiC) single crystal substrate with plurality of trenches and plurality of pillars of silicon carbide in accordance with an example embodiment;

FIG. **9** illustrates a reusable silicon carbide (SiC) single crystal substrate with plurality of trenches partially filled with a carbon layer and plurality of pillars of silicon carbide in accordance with an example embodiment;

FIG. **10** illustrates an epitaxially grown layer with merged epitaxial lateral overgrowth (MELO) with plurality of trenches partially filled with a carbon layer and plurality of pillars of silicon carbide in accordance with an example embodiment;

FIG. **11** illustrates a reusable silicon carbide (SiC) single crystal substrate with plurality of trenches overlying a plurality of micro-voids and plurality of pillars of silicon carbide in accordance with an example embodiment;

FIG. **12** illustrates a reusable silicon carbide (SiC) single crystal substrate with plurality of trenches partially filled with a carbon layer overlying a plurality of micro-voids completely filled with a carbon layer and plurality of pillars of silicon carbide in accordance with an example embodiment;

FIG. **13** illustrates an epitaxially grown layer with merged epitaxial lateral overgrowth (MELO) with plurality of trenches partially filled with a carbon layer overlying a plurality of micro-voids completely filled with a carbon layer and plurality of pillars of silicon carbide in accordance with an example embodiment;

FIG. **14** illustrates an epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) over patterned carbon regions with silicon carbide pillars in silicon carbide substrate in accordance with an example embodiment;

FIG. **15** illustrates a diode contact region in an epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) over patterned carbon regions with silicon carbide pillars in silicon carbide substrate in accordance with an example embodiment;

FIG. **16** illustrates a dielectric isolation layer over diode contact region in epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) over patterned carbon regions with silicon carbide pillars in silicon carbide substrate in accordance with an example embodiment;

FIG. **17** illustrates a patterned dielectric isolation layer over diode contact region in epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) over patterned carbon regions with silicon carbide pillars in silicon carbide substrate in accordance with an example embodiment;

FIG. **18** illustrates patterned metal layer forming one electrode of a Schottky Barrier Diode over patterned dielectric isolation layer over diode contact region in epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) over patterned carbon regions with silicon carbide pillars in silicon carbide substrate in accordance with an example embodiment;

FIG. **19** illustrates a completed Schottky Barrier Diode formed in reusable silicon carbide substrate temporarily bonded to a carrier wafer in accordance with an example embodiment;

FIG. **20** illustrates a completed Schottky Barrier Diode formed in reusable silicon carbide substrate temporarily bonded to a carrier wafer and scanned with a laser to vaporize silicon carbide pillars in reusable silicon carbide in accordance with an example embodiment;

FIG. **21** illustrates a completed Schottky Barrier Diode formed in reusable silicon carbide substrate temporarily bonded to a carrier wafer in an exfoliation tool after scan with a laser to vaporize silicon carbide pillars in reusable silicon carbide substrate in accordance with an example embodiment;

FIG. **22** illustrates exfoliation of Schottky Barrier Diode formed in reusable silicon carbide substrate temporarily bonded to a carrier wafer in an exfoliation tool in accordance with an example embodiment;

FIG. **23** illustrates exfoliated Schottky Barrier Diode formed in reusable silicon carbide substrate temporarily bonded to a carrier wafer in accordance with an example embodiment;

FIG. **24** illustrates exfoliated Schottky Barrier Diode formed in reusable silicon carbide substrate temporarily bonded to a carrier wafer and polished to remove portion of epitaxial layer with patterned carbon regions and plurality of silicon carbide pillars, in accordance with an example embodiment;

FIG. **25** illustrates the back side metallization for an exfoliated Schottky Barrier Diode in reusable silicon carbide substrate temporarily bonded to a carrier wafer and formed in an epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) in accordance with an example embodiment;

FIG. **26** illustrates a completed Schottky Barrier Diode formed in an epitaxially grown layer over an epitaxially grown epitaxial layer with merged epitaxial lateral overgrowth (MELO) exfoliated from a reusable silicon carbide substrate, in accordance with an example embodiment;

FIG. **27** illustrates a portion of reusable silicon carbide substrate after exfoliation in accordance with an example embodiment;

FIG. **28** illustrates a portion of reusable silicon carbide substrate after polishing and reclaim in accordance with an example embodiment;

FIG. **29** shows a block diagram implementation of a semiconductor device using a reusable semiconductor substrate; and

FIG. **30** shows a block diagram implementation of a semiconductor device using a reusable silicon carbide substrate.

DETAILED DESCRIPTION

The following description of embodiment(s) is merely illustrative in nature and is in no way intended to limit the invention, its application, or uses.

For simplicity and clarity of the illustration(s), elements in the figures are not necessarily to scale, are only schematic, are non-limiting, and the same reference numbers in different figures denote the same elements, unless stated otherwise. Additionally, descriptions and details of well-known steps and elements are omitted for simplicity of the description. Notice that once an item is defined in one figure, it may not be discussed or further defined in the following figures.

The terms “first”, “second”, “third” and the like in the Claims or/and in the Detailed Description are used for distinguishing between similar elements and not necessarily for describing a sequence, either temporally, spatially, in ranking or in any other manner. It is to be understood that the terms so used are interchangeable under appropriate circumstances and that the embodiments described herein are capable of operation in other sequences than described or illustrated herein.

Processes, techniques, apparatus, and materials as known by one of ordinary skill in the art may not be discussed in detail but are intended to be part of the enabling description where appropriate.

While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

The current invention is described with an example embodiment of the fabrication of a Schottky Barrier Diode (SBD) using a silicon carbide wafer as the starting substrate. The starting substrate used is a semiconductor substrate that can be used as a reusable semiconductor substrate multiple times for fabrication of semiconductor devices. Alternatively, other devices such as transistors, passive devices, or power transistors can be formed using the described process flow. While silicon carbide substrate is used in the example embodiment, the invention can be implemented in other semiconductor substrates such as gallium nitride, gallium arsenide, indium phosphide, silicon, silicon on insulator (SOI) among others. In addition, the invention may be used in other semiconductor devices such as photonic devices, lasers, light emitting diodes, RF devices, among others.

FIG. **1** is an illustration of a reusable silicon carbide substrate **100** in accordance with an example embodiment. Silicon carbide substrate **100** is used as a starting material for the fabrication of the Schottky Barrier Diode. In one embodiment, silicon carbide substrate **100** is a crystalline 4H silicon carbide wafer with a preferred crystalline orientation of <0001>with an offcut towards<1120> of 4 degrees. In one embodiment, a thickness of silicon carbide substrate **100** is in the range of 300-400 microns. In one embodiment, silicon carbide substrate **100** may be a single side polished or double side polished wafer and can be considered as the parent wafer, for considerations that are described in subsequent process steps in the implementation of the current invention. In one embodiment, silicon carbide substrate **100** is the basic platform on which the example embodiment is implemented to support the process flow in accordance with the current invention. Silicon carbide substrate **100** is a reusable semiconductor substrate that is used for fabrication of semiconductor devices multiple times on the same substrate, in accordance with the current invention.

FIG. **2** is an illustration of a layer **200** formed overlying silicon carbide substrate **100** in accordance with an example embodiment. In one embodiment layer **200** comprises a carbon material and is formed on a surface of silicon carbide substrate **100**. In one embodiment, layer **200** is formed on the surface of silicon carbide substrate **100** using a variety of methods. In one embodiment, layer **200** is formed on the surface of silicon carbide substrate **100** using sputtered carbon. In another embodiment, layer **200** is deposited using CVD (Chemical Vapor Deposition) or PVD (Physical Vapor Deposition) among other techniques.

In another embodiment, layer **200** is formed on the surface of silicon carbide substrate **100** by depositing a polymer layer and then converting the polymer layer into a carbon layer using pyrolysis. In one embodiment, the polymer layer is composed of Parylene. Parylene (trade name for poly p-xylylene) is a semicrystalline thermoplastic polymer deposited using CVD (Chemical Vapor Deposition). Parylene C is one version of parylene which is a chlorinated poly para-xylylene polymer and is deposited using CVD (Chemical Vapor Deposition) to form a conformal coating. Parylene C deposition consists of heating a solid, granular material called dimer under vacuum to vaporize into a dimeric gas in a temperature range of (100-150) ° C. The dimeric gas is then pyrolyzed to cleave the dimer into its monomeric form. The monomer gas is then used in a vacuum chamber at room temperature to deposit conformally on all surfaces of the samples inside a vacuum chamber as a polymer film. In one embodiment, silicon carbide substrate **100** is coated with polymer layer of Parylene C conformally. The thickness of Parylene C is in a range of 500 nanometers (nm) to several micrometers (um). The deposited Parylene C is converted into layer **200** which is a carbon layer using a process of pyrolysis. The pyrolysis of Parylene C into carbon is done in an inert environment. The pyrolysis of Parylene C converts the Parylene C into a carbon layer which may be amorphous or polycrystalline. In one embodiment, the temperature for pyrolysis is between (600-1200) ° C. and the inert environment is nitrogen or a forming gas (nitrogen and hydrogen) among others. In one embodiment, to account for the shrinkage of the Parylene C during the pyrolysis process, multiple layers of Parylene C are deposited and converted to carbonized layer to achieve the target thickness of carbon layer **200**.

In one embodiment, the polymer layer is composed of photoresist. The photoresist used as the polymer may be of positive or negative polarity. In one embodiment, polymer photoresist layer is spin coated on the surface of silicon carbide substrate **100**. Photoresist polymer layer may also be spray coated on surface of silicon carbide substrate **100**. In one embodiment, after the deposition of the polymer photoresist it is soft baked to drive out solvents. Soft baking polymer photoresist means that it is heated to a temperature in the range of (90-100) ° C. in an inert environment such as nitrogen to drive out solvents. Multiple layers of polymer photoresist may be used to achieve the desired thickness of polymer photoresist.

Polymer photoresist layer deposited on surface of silicon carbide substrate **100** is converted to carbon layer **200** by process of pyrolysis. Pyrolysis of polymer photoresist layer consists of thermal treatment in an inert environment to form carbon layer **200**. Pyrolysis of polymer photoresist layer into carbon layer **200** can comprise multiple intermediate thermal treatments. In one embodiment, polymer photoresist layer is baked in nitrogen environment at 90° C. (typically called a soft bake), followed by bake at 115° C. (typically called a hard bake) in the nitrogen environment. Hard baked polymer photoresist layer is then cured at 450° C. in the nitrogen environment and then pyrolyzed in a furnace in the nitrogen environment at (800-1200) ° C. to convert the polymer photoresist layer to carbon layer **200**. In another embodiment, a forming gas (nitrogen and hydrogen) is used for the pyrolysis of polymer photoresist layer to carbon layer **200**.

FIG. **3** is an illustration of a hard mask layer **300** deposited overlying carbon layer **200** on silicon carbide substrate **100** in accordance with an example embodiment. In one embodiment, hard mask layer **300** is deposited on a surface of carbon layer **200**. Hard mask layer **300** is deposited using techniques such as CVD (Chemical Vapor Deposition), LPCVD (low pressure chemical vapor deposition), PECVD (Plasma Enhanced Chemical Vapor Deposition), APCVD (Atmospheric Pressure Chemical Vapor Deposition), SACVD (Sub Atmospheric Chemical Vapor Deposition), PVD (Physical Vapor Deposition), or ALD (Atomic layer Deposition). In the example implementation, hard mask layer **300** is composed of Al_2O_3 deposited using ALD (Atomic Layer Deposition) The thickness of hard mask layer **300** is determined by the specific requirements of the implementation and is well known to those skilled in the art and is the range of 20-300 nm.

FIG. **4** is an illustration of a plurality of openings **400** formed in hard mask layer **300** of FIG. **3** in accordance with an example embodiment. In one embodiment, the process steps disclosed herein below will lead to the formation of a device in substrate **100** and more specifically a Schottky Barrier Diode as an example semiconductor device. Hard mask layer **300** of FIG. **3** deposited over the carbon layer **200** over surface of substrate **100** is patterned to subsequently support the formation of plurality of openings **400**. In one embodiment, the surface of carbon layer **200** is exposed by removal of hard mask layer **300** to form plurality of openings **400**. Plurality of openings **400** are formed in hard mask layer **300** using methods of lithography and etching common to the semiconductor industry. Patterned hard mask **405** is left in areas to protect portions of carbon layer **200** from being etched. The shape of plurality of openings **400** are determined by the requirements of epitaxial growth in subsequent steps in the implementation of the example embodiment. In one embodiment, plurality of openings **400** may be in the shape of squares or rectangles. In another embodiment, plurality of openings **400** may be in the shape of triangles, hexagons or diamonds. In another embodiment, plurality of openings **400** may be in the shape of stripes which may be horizontal, vertical, or sloped at a diagonal angle. In one embodiment, the size of plurality of openings **400** may be in the range of (20-500) nm and determined by the requirements of epitaxial overgrowth in the subsequent steps of fabrication of the example device. In one embodiment, spacing between adjacent openings of plurality of openings **400** is determined by the requirements of epitaxial overgrowth in the subsequent steps of fabrication of the example device and can be in the range of 500 nm to 5 micrometers. Plurality of openings **400** are generated on a surface of hard mask layer **300** of FIG. **3** by using lithography techniques that are well known to those skilled in the art. In one embodiment, plurality of openings **400** are implemented using optical lithography using UV, DUV or EUV. In another embodiment, plurality of openings **400** are implemented using an electron beam direct write technique.

In yet another embodiment, plurality of openings **400** are implemented using Nano-Imprint Lithography (NIL).

In one example embodiment, plurality of openings **400** are implemented by first coating a surface of hard mask layer **300** with a photosensitive layer of photoresist, which may be positive or negative in its chemistry. In the example embodiment, positive photoresist is used in coating the surface of hard mask layer **300**. An optical tool called a stepper is used to transfer the pattern of openings on to the positive photoresist layer using chemistries that are well known to those skilled in the art. The choice of the photoresist layer, thickness of the photoresist layer, the exposure and develop times for the subsequent chemical steps are well known to those skilled in the art and determined by the requirements of accurate pattern transfer from the photoresist layer to hard mask layer **300** to subsequently form plurality of openings **400**. The stepper transfers the pattern of plurality of openings **400** to cover the surface of hard mask layer **300** over carbon layer **200** over silicon carbide substrate **100**.

After the pattern transfer is completed using lithography, the next step is the patterning of hard mask layer **300** using etching techniques to selectively remove the hard mask layer **300** of FIG. **3** overlying carbon layer **200** on silicon carbide substrate **100** leaving patterned hard mask **405** on carbon layer **200** over substrate **100**. The selective removal of hard mask layer **300** to form patterned hard mask **405** may use Reactive Ion Etching (RIE). Different gases may be used to form a plasma to selectively remove the portions of hard mask layer **300** exposed by the patterned photoresist. The choice of gases for the RIE is determined by hard mask layer **300** used in the implementation. In the example embodiment, with Al_2O_3 layer used as hard mask layer **300**, fluorine-based chemistries such as SF_6, CF_4, CHF_3, and other gases may be used in the RIE. Accordingly, in the example embodiment with Al_2O_3 layer as hard mask layer **300**, plurality of openings **400** are etched in hard mask layer **300** using a fluorine-based chemistry that exposes the surface of carbon layer **200** in plurality of openings **400**. Patterned hard mask **405** remains in areas overlying carbon layer **200** to protect or mask the surface of carbon layer **200** from etching.

FIG. **5** is an illustration of a plurality of trenches **500** formed in carbon layer **200** in accordance with an example embodiment. Plurality of trenches **500** expose a surface of silicon carbide substrate **100** after etching carbon layer **200** exposed by plurality of openings **400** from FIG. **4**. Thus, the exposed surface of carbon layer **200** over silicon carbide substrate **100** in plurality of openings **400** is etched to form plurality of trenches **500** in FIG. **5** using RIE (Reactive Ion Etching). In one embodiment, carbon layer **200** is etched using patterned hard mask **405** to form plurality of trenches **500** with an aspect ratio that is determined by the requirements of epitaxial growth in subsequent processing of the example device. Thus, plurality of openings **400** are etched in the carbon layer **200** to expose the surface of silicon carbide substrate **100** to form plurality of trenches **500**. The depth of plurality of trenches **500** is dependent on the thickness of carbon layer **200** and may be in the range of 500 nm to 3 micrometers. An oxygen plasma with fluorine chemistry or argon may be used for etching carbon layer **200** using patterned hard mask **405**. An inductively coupled plasma (ICP) with high density may also be used to form plurality of trenches **500** using patterned hard mask **405** over carbon layer **200** over silicon carbide substrate **100**. After plurality of trenches **500** are formed by etching the exposed surfaces of plurality of openings **400** in carbon layer **200** over silicon carbide substrate **100**, the photoresist is removed using resist stripping techniques well known to those skilled in the art. The resist stripping technique may use dry etching or wet etching or a combination of dry and wet stripping techniques.

FIG. **6** is an illustration of an example embodiment where patterned hard mask **405** is removed forming a plurality of trenches **500** in accordance with an example embodiment. FIG. **6** further illustrates plurality of patterned carbon regions **600** formed overlying silicon carbide substrate **100**. Patterned hard mask **405** of FIG. **5** is removed using wet etching, dry etching or a combination of wet etching and dry etching, depending on hard mask layer **300**. In an example embodiment, hard mask layer **300** of FIG. **3** comprises Al_2O_3 that is etched using dilute BOE (Buffered Oxide Etch). Silicon carbide substrate **100** with plurality of patterned carbon regions **600** and plurality of trenches **500** is then cleaned in preparation for the next step in the fabrication of the example device. In one embodiment, the pattern of plurality of trenches **500** are shaped as triangles or hexagons to expose (**1120**) or equivalent crystal planes since these orientations facilitate high quality epitaxial overgrowth with low defect density in subsequent processing steps in accordance with the current invention.

FIG. **7** is an illustration of an epitaxial layer **700** formed overlying the surface of patterned carbon regions **600** in accordance with an example embodiment. In one embodiment, epitaxial layer **700** is a silicon carbide epitaxial layer formed over patterned carbon regions **600** and coupling to silicon carbide substrate **100**. Epitaxial layer **700** may be a N+ heavily doped layer forming a buffer epitaxial layer. In one embodiment, epitaxial layer **700** forms a plurality of silicon carbide pillars **710** between the patterned carbon regions **600** by epitaxial growth of silicon carbide from surface of silicon carbide substrate **100** in plurality of openings **500** from FIG. **6**.

In general, epitaxial layer **700** is formed on silicon carbide substrate **100** with patterned carbon regions **600** overlying silicon carbide substrate **100** in a silicon carbide epitaxial reactor. In the epitaxial reactor, silicon carbide grows with the crystalline orientation of exposed silicon carbide substrate **100** in plurality of openings **500** from FIG. **6**. In the epitaxial reactor, silicon carbide grows vertically as well as laterally with patterned carbon regions **600** inhibiting the epitaxial growth with growth conditions that are well known to those skilled in the art. The silicon carbide forms array of silicon carbide pillars **710** due to the vertical growth of silicon carbide and then forms epitaxial layer **700** when it grows laterally. Note that silicon carbide pillars **710** corresponds to silicon carbide between adjacent patterned carbon regions **600**. Once above patterned carbon regions **600**, the silicon carbide grows as a uniform layer.

The lateral fronts of the epitaxial regions of silicon carbide epitaxial layer **700** merge due to epitaxial lateral overgrowth (ELO) or merged epitaxial lateral overgrowth (MELO). The process of epitaxial crystal growth is used to form a single crystal layer of silicon carbide over patterned carbon regions **600** forming epitaxial layer **700** with silicon carbide pillars **710**. In the example embodiment, epitaxial layer **700** is an epitaxial layer of silicon carbide. This method of ELO or MELO over the regions of patterned carbon **600** enables the formation of epitaxial layer **700** with low defect density which is mechanically supported by patterned carbon regions **600** and plurality of silicon carbide pillars **710**. Silicon carbide substrate **100** with patterned carbon regions **600** and plurality of silicon carbide pillars **710** below epitaxial layer **700** forms a plane where epitaxial layer **700** can be exfoliated from substrate **100** in subsequent process steps. The patterned carbon regions **600** along with plurality of pillars **710** enables the formation of epitaxial layer **700** as a single crystal silicon carbide layer by ELO or MELO. This plane of separation comprises patterned carbon regions **600** and plurality of pillars **710**. In one embodiment, epitaxial layer **700** is grown in an epitaxial reactor using CVD (Chemical Vapor Deposition) epitaxial growth processes or by modified bulk crystal growth processes such as high Temperature CVD or by Physical Vapor Transport (PVT). The exfoliation process of separating substrate **100** from epitaxial layer **700** will be disclosed in detail herein below.

Patterned carbon regions **600** is compatible with the epitaxial growth process since carbon is incorporated in the silicon carbide crystalline structure during the epitaxial growth where gases such as acetylene (C_2H_2) is used in the epitaxial reactor along with other process gases such as DCS (Dichlorosilane), TCS (trichlorosilane), silane among other process gases. Patterned carbon regions **600** may be amorphous or polycrystalline depending on the method of forming the carbon layer **200** from FIG. **2** and is capable of withstanding high temperature processing of the silicon carbide devices formed above epitaxial layer **700** in subsequent process steps. In addition, carbon has a Young's modulus of 70 GPa compared to 700 GPa of singe crystal silicon carbide.

By appropriate design of mechanical and thermal consideration of patterned carbon regions **600** with plurality of silicon carbide pillars **710**, the forces required for exfoliation of single crystal silicon carbide epitaxial layer **700** may be tailored to be optimized such that the entire structure can withstand the thermal and mechanical processes during subsequent device formation steps while also being able to be separated by the exfoliation process in the plane of the patterned carbon regions **600** with plurality of silicon carbide pillars **710**. In the example embodiment, epitaxial layer **700** comprises of N+4H Silicon Carbide and can be of a thickness of about 5-20 micrometers. In another embodiment, P+ silicon carbide can be used for epitaxial layer **700**. The doping of epitaxial layer **700** is high enough to provide an ohmic contact for the silicon carbide device formed on epitaxial layer **700** during subsequent processing steps. Patterned carbon regions **600** inhibits the growth of silicon carbide epitaxial layer over it while enabling the lateral growth of silicon carbide from plurality of silicon carbide pillars **710** that grow from surface of reusable silicon carbide substrate **100** and also forms a portion of the plane where exfoliation is initiated in a later stage of the invention, as described in more detail in subsequent processing steps.

FIG. **8** is an illustration of silicon carbide substrate **100** patterned to form a plurality of trenches **800** and a plurality of silicon carbide pillars **810** in accordance with an alternate example embodiment. In one embodiment, pattern of plurality of trenches **800** is formed by coating silicon carbide substrate with a hard mask layer for patterning and etching. The hard mask layer used for patterning of trenches **800** may be composed of one or more layers. In one embodiment, the hard mask layer is composed of LPCVD Silicon Nitride coated with LPCVD silicon oxide. The hard mask layers are patterned using photoresist and optical lithography and etched using reactive ion etching (RIE). In one embodiment, hard mask layers of silicon nitride and silicon oxide are patterned using RIE using fluorine chemistry. After the patterning and etching of the hard mask layers, silicon carbide substrate **100** is etched using Reactive Ion Etching using a fluorine chemistry to form plurality of trenches **800** and leaving plurality of silicon carbide pillars **810**. Inductively Coupled Plasma (ICP) may also be used to etch silicon carbide substrate **100**. After etching of plurality of trenches **800** and formation of plurality of silicon carbide pillars **810**, hard mask layers are stripped using wet etching, dry etching or a combination of wet and dry etching. The aspect ratio of the plurality of pillars of silicon carbide may be between 2-10.

FIG. **9** is an illustration of silicon carbide substrate **100** patterned with plurality of trenches **800** filled with a carbon layer **900** partially filling plurality of trenches **800** in accordance with the alternate embodiment. Carbon layer **900** partially filling plurality of trenches **800** can be formed by sputter deposition of a carbon film on the surface of silicon carbide substrate **100** patterned with a plurality of trenches **800** and partially filled with carbon layer **900** by blanket etching of carbon film deposited on surface of silicon carbide substrate **100**. Carbon layer **900** is removed from surfaces of plurality of silicon carbide pillars **810**. In one embodiment, carbon layer **900** may be formed by deposition of a polymer on surface of silicon carbide substrate **100**, etching back from top of plurality of silicon carbide pillars **810** and pyrolysis of polymer layer to form carbon layer **900** partially filling plurality of trenches **800**. Polymer layer used for pyrolysis may be deposited by CVD (Chemical Vapor Deposition), PVD (Physical Vapor Deposition), spin coating, spray coating among other techniques.

FIG. **10** is an illustration of an epitaxial layer **1000** formed over silicon carbide substrate **100** in accordance with the alternate embodiment. In one embodiment, epitaxial layer **1000** is a silicon carbide epitaxial layer forming a buffer epitaxial layer with low defect density. Epitaxial layer **1000** is formed by placing silicon carbide substrate **100** with plurality of trenches **800** from FIG. **9** partially filled with carbon layer **900** in a silicon carbide epitaxial reactor with epitaxial growth of silicon carbide using ELO (Epitaxial Lateral Overgrowth) or MELO (Merged Epitaxial Lateral Overgrowth) to form a single crystal epitaxial layer **1000**. In the example, single crystal epitaxial layer **1000** overlies silicon carbide substrate **100**, carbon layer **900**, and plurality of pillars **810**.

FIG. **11** is an illustration of silicon carbide substrate **100** patterned with a plurality of trenches **1100** having a plurality of micro-voids **1110** in accordance with an alternate embodiment. In one embodiment, a pattern corresponding to plurality of trenches **1100** is formed by coating silicon carbide substrate with a hard mask layer for patterning and etching. Hard mask layer used for patterning of plurality of trenches **1100** may be composed of one or more layers. In one embodiment, the hard mask layer is composed of LPCVD Silicon Nitride. The hard mask layer is patterned using photoresist and optical lithography and etched using reactive ion etching (RIE). In one embodiment, the hard mask layer of silicon nitride is patterned using RIE with a fluorine chemistry. After the patterning and etching of the hard mask layers, silicon carbide substrate **100** is etched using Reactive Ion Etching with the fluorine chemistry to form plurality of trenches **1100**. Inductively Coupled Plasma (ICP) may also be used to etch silicon carbide substrate **100**. After etching of plurality of trenches **1100**, a conformal layer of a spacer layer is deposited and etched to form spacers in the walls of trenches **1100**. The plurality of micro-voids **1110** are then etched in silicon carbide substrate **100** using isotopic etching chemistry below plurality of trenches **1100**. In one embodiment, plurality of micro-voids are wider than the plurality of trenches **1100** and form plurality of pillars **1120**. Plurality of pillars **1120** correspond to the silicon carbide between adjacent micro-voids of plurality of micro-voids **1110**. After forming of trenches **1100** with plurality of micro-voids **1110** below plurality of trenches **1100** and leaving plurality of pillars **1120** in silicon carbide substrate **100**, the hard mask layer and spacer layer is removed using wet chemistry. If the hard mask layer and spacer layer is silicon nitride, hot phosphoric acid is used for the removal of hard mask layer and spacer layer.

FIG. **12** is an illustration of silicon carbide substrate **100** patterned with a plurality of trenches **1100** over a plurality of micro-voids **1110** from FIG. **11** filled with a carbon layer **1200** partially filling plurality of trenches **1100** and completely filling plurality of micro-voids **1110** in accordance with the alternate embodiment. Carbon layer **1200** partially filling plurality of trenches **1100** can be formed by sputter deposition of a carbon film on the surface of silicon carbide substrate **100** patterned with a plurality of trenches **1100** and plurality of micro-voids **1110** and blanket etching of carbon film deposited on the surface of silicon carbide substrate **100** with plurality of trenches **1100**. In one embodiment, carbon layer **1200** may be formed by deposition of a polymer on the surface of silicon carbide substrate **100**, etching back from a top of plurality of silicon carbide pillars **1120**, and pyrolysis of the polymer layer to form carbon layer **1200** partially filling plurality of trenches **1100** and completely filling plurality of micro-voids **1110**. Polymer layer used for pyrolysis to form carbon layer **1200** may be deposited by CVD (Chemical Vapor Deposition), PVD (Physical Vapor Deposition), spin coating, spray coating among other techniques.

FIG. **13** is an illustration of an epitaxial layer **1300** over silicon carbide substrate **100** patterned in accordance with the alternate embodiment. In one embodiment, epitaxial layer **1300** is a silicon carbide epitaxial layer forming a buffer epitaxial layer with low defect density. Epitaxial layer **1300** is formed by placing silicon carbide substrate **100** with plurality of trenches **1100** from FIG. **11** partially filled with carbon layer **1200** over a plurality of micro-voids **1110** completely filled with carbon layer **1200** in a silicon carbide epitaxial reactor with epitaxial growth of silicon carbide using ELO (Epitaxial Lateral Overgrowth) or MELO (Merged Epitaxial Lateral Overgrowth) to form a single crystal epitaxial layer **1300** with patterned carbon layer **1200** with a plurality of silicon carbide pillars **1120**.

FIG. **14** is an illustration of an epitaxial layer **1400** formed overlying epitaxial layer **700** in accordance with an example embodiment. In one embodiment, a device is formed in epitaxial layer **1400** that is grown overlying epitaxial layer **700** in an epitaxial reactor. In one embodiment, epitaxial layer **1400** comprises silicon carbide. In an example embodiment, the device that is formed in epitaxial layer **1400** is a silicon carbide device that is formed in subsequent processing steps. In one embodiment, prior to the epitaxial growth of epitaxial layer **1400**, a surface of epitaxial layer **700** may be lightly polished using a polishing step called kiss polish to remove any surface defects on the surface of epitaxial layer **700**. In one embodiment, the doping and thickness of device epitaxial layer **1400** are determined by the electrical requirements of devices that are formed in device epitaxial layer **1400**. In one embodiment, the thickness of device epitaxial layer **1400** is determined by a breakdown voltage of the device formed in the epitaxial layer **1400** in subsequent processing steps and is typically between 10-30 micrometers. In the example embodiment, epitaxial layer **1400** is doped N- and has a thickness of about 10-12 micrometers for a device breakdown voltage of 1200 Volts. Epitaxial layer **1400** formed overlying epitaxial layer **700** is used for formation of silicon carbide devices using processes well known to those skilled in the art. In the example embodiment, epitaxial layer **1400** is used for formation of a Schottky Barrier Diode in accordance with the current invention. It should be noted that epitaxial layer **1400** can be formed overlying any of the different embodiments comprising the ELO and MELO epitaxial layer disclosed herein. Also, any subsequent processing steps can also be performed in all the different embodiments disclosed herein.

In one embodiment, device epitaxial layer **1400** overlying epitaxial layer **700** enables the formation of silicon carbide devices that can subsequently be separated from silicon carbide substrate **100** by method of an exfoliation process that may be thermal, mechanical, and other techniques. A combination of techniques may also be used in the exfoliation process of device epitaxial layer **1400** and epitaxial layer **700** on which semiconductor devices can be fabricated. It should be also noted that the exfoliation process disclosed herein supports reuse of silicon carbide substrate **100** as epitaxial layer **1400** comprises only a portion of silicon carbide substrate **100**. In one embodiment, a surface of silicon carbide substrate **100** can be prepared to be reused to form more devices.

In one embodiment, device epitaxial layer **1400** overlying epitaxial layer **700** may be grown to a thickness of (150-400) microns to enables the formation of silicon carbide substrate that can subsequently be separated from silicon carbide substrate **100** by method of an exfoliation process thereby enabling formation of kerfless silicon carbide substrate. In one embodiment, device epitaxial layer **1400** may be very lightly doped to act as a semi-insulating substrate.

FIG. **15** is an illustration of epitaxial layer **1400** being doped to lower resistivity in accordance with an example embodiment. To reduce a contact resistance of the device, dopants are implanted on a surface of device epitaxial layer **1400**. In one embodiment, a dopant species, dose, energy and other parameters are determined by the design of the Schottky Barrier Diode. In one embodiment, the implanted layer is N+. The implanted dopants are then subsequently annealed to form an ohmic contact region **1500**.

FIG. **16** is an illustration of a dielectric isolation layer **1600** deposited on ohmic contact region **1500** on epitaxial layer **1400** in accordance with an example embodiment. Dielectric isolation layer **1600** is deposited by using PECVD Silicon Dioxide, PECVD Silicon Nitride, PECVD, or Silicon Oxynitride among other films. In one embodiment, a thickness of dielectric isolation layer **1600** is in a range of (1-4) micrometers. In the example embodiment, dielectric isolation layer **1600** is PECVD Silicon Oxide and is approximately one micrometer thick.

FIG. **17** is an illustration of dielectric isolation layer **1600** being patterned in accordance with an example embodiment. Dielectric isolation layer **1600** is patterned and etched using standard wafer processing steps to form contact openings **1700** exposing portion of ohmic contact region **1500** in epitaxial layer **1400**. In one embodiment, patterning is done using photolithography techniques and etching of dielectric isolation layer **1600** to form contact openings **1700** is done using RIE (Reactive Ion Etching), wet etching or a combination of etching steps. In the example embodiment, contact openings **1700** are patterned using RIE.

FIG. **18** is an illustration of a metal contact layer **1800** configured to form an electrode of the Schottky Diode in accordance with an example embodiment. In one embodiment, contact openings **1700** from FIG. **17** are covered with metal contact layer **1800**. Metal contact layer **1800** is deposited using sputtering, e-beam evaporation, electrodeposition among other techniques and can also use a combination of metal deposition techniques. Metal contact layer **1800** may be patterned using lithography and etched. In addition, lift-off techniques may also be used for the deposition and patterning of metal contact layer **1800**, as will be evident to those skilled in the art. Metal contact layer **1800** may be annealed or sintered to ensure good ohmic contact with ohmic contact region **1500** from FIG. **15**. After formation of metal contact layer **1800**, a passivation layer may be deposited and patterned to expose bond pads of the example device in accordance with the current invention. At this stage of the example embodiment, the fabrication of a semiconductor device such as the Schottky Barrier Diode is complete in epitaxial layer **1400** formed over epitaxial layer **700** which overlies patterned carbon layer **600**, carbon layer **900** of FIG. **9**, or carbon layer **1200** of FIG. **12**. In an example embodiment, front side metallization results in silicon carbide substrate **100** with Schottky Barrier Diode 1950.

FIG. **19** is an illustration of a carrier wafer **1900** temporarily coupled to reusable silicon carbide substrate **100** with Schottky Barrier Diode 1950 in accordance with an example embodiment. In general, carrier wafer **1900** is a substrate used for handling epitaxial layer **700** and epitaxial layer **1400**. Silicon carbide substrate **100** with Schottky Barrier Diode 1950 is temporarily coupled to carrier wafer **1900** to enable an exfoliation process. The exfoliation process occurs at an exfoliation layer comprising patterned carbon region **600** adjacent to plurality of silicon carbide pillars **710**. Thus, patterned carbon region **600** and plurality of silicon carbide pillars **710** form a layer on a plane where separation occurs during the exfoliation process.

In one embodiment, a plane of the exfoliation layer is substantially parallel to the surface of substrate **100**. In one embodiment, silicon carbide substrate **100** with completed Schottky Barrier Diode 1950 is attached to carrier wafer **1900** by adhesives such as UV sensitive glue among others. In the example, epitaxial layer **1400** and epitaxial layer **700** are coupled between silicon carbide substrate **100** and carrier wafer **1900**. Carrier wafer **1900** may be borosilicate glass which is UV transparent and may be used with a UV curable adhesive for the bonding. Different methods of exfoliation may be used to separate semiconductor devices formed in device epitaxial layer **1400** overlying epitaxial layer **700** coupled to carrier wafer **1900** along the plane of the exfoliation layer. As an example, the exfoliation process is achieved by using an electrostatic chuck to hold the assembly of silicon carbide substrate **100** with Schottky Barrier Diode 1950 and carrier wafer **1900** and applying normal and shear stresses to fracture the exfoliation layer comprising patterned carbon region **600** and plurality of pillars **710**. In another example, the exfoliation process is done using thermal stresses to initiate fracture of the exfoliation layer comprising patterned carbon region **600** and plurality of pillars **710**. A combination of techniques may also be used for the exfoliation process of silicon carbide substrate **100** with device epitaxial layer **1400** and epitaxial layer **700**. Note that the exfoliation process examples herein above would also work for the exfoliation layer comprising carbonized layer **900** with plurality of pillars **810** of FIG. **10** or carbonized layer **1200** of FIG. **13** filling plurality of micro-voids **1110** in FIG. **11** with plurality of pillars **1120** of FIG. **13**.

FIG. **20** is an illustration of a reusable silicon carbide substrate **100** with Schottky Barrier Diode 1950 temporarily coupled to a carrier wafer **1900** undergoing an exfoliation process using a laser **2030** in accordance with an example embodiment. Reusable silicon carbide substrate **100** with Schottky Barrier Diode 1950 temporarily coupled to a carrier wafer **1900** is placed above a laser **2030** such that a laser beam **2020** is scanned into the exfoliating layer comprising patterned carbon region **600** and plurality of silicon carbide pillars **710** from FIG. **19**. The wavelength of the laser **2030** is chosen so that it is substantially transparent to reusable silicon carbide substrate **100** and couples the laser energy to the patterned carbon regions **600** from FIG. **19**. The laser energy is selectively absorbed by the patterned carbon layer and can reach several thousand degrees of temperature in Celsius. The laser **2030** may be used in continuous or pulsed mode. In one embodiment, laser **2030** is used in pulsed mode so that the energy coupled converts patterned carbon regions **600** to be vaporized or partially vaporized carbonized regions **2000**. The energy coupled to the patterned carbon regions **600** from FIG. **19** causes the plurality of silicon carbide pillars **710** in FIG. **19** to be vaporized or partially vaporized thereby forming weak regions of vaporized silicon carbide **2010**. By scanning laser beam **2020** of laser **2030**, the plane of patterned carbon region **600** and plurality of silicon carbide pillars **710** from FIG. **19** is converted into the exfoliating layer comprising vaporized or partially vaporized silicon carbide **2010** and vaporized or partially vaporized carbonized regions **2000**. This exfoliating layer weakly couples to Schottky Barrier Diode 1950 formed in epitaxial layer **700** and epitaxial layer **1400** to reusable silicon carbide substrate **100**. Laser **2030** may have a wavelength of 532 nanometers, 1064 nanometers, 623-700 nanometers or 632 nanometers for exfoliation of reusable silicon carbide substrate **100**. For Gallium Nitride the appropriate wavelengths may be between 400-1000 nanometers. For silicon substrates, the appropriate wavelengths may be in the UV (Ultra-Violet) range of 350 nanometers.0. In one embodiment, the laser exfoliation process may be used prior to the device fabrication steps as described in FIG. **15-18**. In this embodiment, the plane of patterned carbon region **600** and plurality of silicon carbide pillars **710** from FIG. **19** is converted into the exfoliating layer comprising vaporized or partially vaporized silicon carbide **2010** and vaporized or partially vaporized carbonized regions **2000** prior to any device formation. This exfoliating layer weakly couples the portion of substrate formed in epitaxial layer **700** and epitaxial layer **1400** to reusable silicon carbide substrate **100** prior to any device fabrication. This exfoliating layer enables the portion of substrate formed in epitaxial layer **700** and epitaxial layer **1400** to be partially released from reusable silicon carbide substrate **100** enabling easier exfoliation in a subsequent step after the device fabrication is completed. Laser **2030** may be used to optimize energy coupled to patterned carbon layer by varying the power, pulse width, pulse duration among other parameters. Laser **2030** may be operated to reduce heating of epitaxial layer **700** and epitaxial layer **1400** with a sharp fall off due to selective energy coupling to patterned carbon layer **600** from FIG. **19**. By scanning laser **2030** across the entire surface of reusable silicon carbide substrate **100**, the exfoliating process produces an exfoliation layer that weakly couples reusable silicon carbide substrate **100** to Schottky Barrier Diode 1950 formed in epitaxial layer **700** and epitaxial layer **1400**.

FIG. **21** is an illustration of reusable silicon carbide substrate **100** being separated from Schottky Barrier Diode 1950 in accordance with an example embodiment. As shown, Schottky Barrier Diode 1950 is temporarily coupled between carrier wafer **1900** and reusable silicon carbide substrate **100** by the exfoliation layer comprising vaporized or partially vaporized silicon carbide **2010** and vaporized or partially vaporized carbonized regions **2000**.

In one embodiment, reusable silicon carbide substrate **100** with Schottky Barrier Diode 1950 is temporarily coupled to carrier wafer **1900** with exfoliation layer comprising vaporized or partially vaporized silicon carbide **2010** and vaporized or partially carbonized regions **2000** is placed in an exfoliating tool consisting of an upper portion **2100** and a lower portion **2110**. A surface of the assembly of carrier wafer **1900** is coupled to upper portion **2100** of the exfoliating tool. Carrier wafer **1900** also couples to Schottky Barrier Diode 1950. The lower portion **2110** of the exfoliating tool is coupled to a surface of reusable silicon carbide substrate **100**. Coupling to upper portion **2100** and lower portion **2110** of the exfoliating tool comprises an adhesive layer, UV tape, electrostatic chuck among other methods of coupling.

Exfoliating tool consisting of an upper portion **2100** and a lower portion **2110** to which the assembly of carrier **1900** and Schottky Barrier Diode 1950 and reusable silicon carbide substrate **100** is coupled is capable of exerting pulling forces normal to the surface of the assembly of carrier wafer **1900** and Schottky Barrier Diode 1950 formed overlying reusable silicon carbide substrate **100** and also rotational forces due to torque parallel to surface of carrier wafer **1900** and Schottky Barrier Diode 1950 formed overlying reusable silicon carbide substrate **100**.

In one embodiment, pulling force **2150** exerted by upper portion **2100** normal to the surface of the assembly of carrier wafer **1900** and Schottky Barrier Diode 1950 formed overlying reusable silicon carbide substrate **100** may be opposite to pulling force **2130** exerted by lower portion **2110** of exfoliating tool. Rotation force imparted by upper portion **2100** produces torque **2140** parallel to surface of carrier wafer **1900** and Schottky Barrier Diode 1950 formed overlying reusable silicon carbide substrate **100** and may be of opposite direction to torque **2120** produced by lower portion **2110**. Pulling force **2150** and **2130** and torque **2140** and **2120** can be computer controlled with sensors providing a feedback mechanism to separately and simultaneously control the pulling forces and shear forces exerted on assembly of carrier wafer **1900** and Schottky Barrier Diode 1950 formed overlying reusable silicon carbide substrate **100**. Alternatively, a rotation force can be applied to one of carrier wafer **1900** or reusable silicon carbide substrate **100** instead of both. Similarly, a pulling force can be applied to one of carrier wafer **1900** or reusable silicon carbide substrate **100** instead of both.

FIG. **22** is an illustration of a portion of silicon carbide substrate **2200** with Schottky Barrier Diode 1950 coupled to carrier wafer **1900** after exfoliation in an exfoliation tool in accordance with an example embodiment. Thus, a silicon carbide substrate **2200** is formed after exfoliation along a fracture plane **2220** and separated from a remaining silicon carbide substrate **2230** after the exfoliation process. In the example embodiment, silicon carbide substrate **2200** comprises epitaxial layer **700** and epitaxial layer **1400** both formed of silicon carbide. In one embodiment, assembly of completed Schottky Barrier Diode 1950 fabricated in device epitaxial layer **1400** over epitaxial layer **700** and temporarily coupled to carrier wafer **1900** is exfoliated from remaining silicon carbide substrate **2230** along the plane comprising vaporized or partially vaporized silicon carbide **2010** and vaporized or partially vaporized carbonized regions **2000** from FIG. **21**. FIG. **22** is not drawn to scale since thickness of remaining silicon carbide substrate **2230** is in the range of 300-400 micrometers, while the portion of silicon carbide substrate **2200** is in the range of 20-60 micrometers.

FIG. **23** is an illustration of a portion of silicon carbide substrate **2200** with Schottky Barrier Diode 1950 coupled to carrier wafer **1900** after removal from the exfoliation tool in accordance with an example embodiment. Thus, a silicon carbide substrate **2200** is formed after exfoliation along a fracture plane **2220** and separated from a remaining silicon carbide substrate **2230** after the exfoliation process. In the example embodiment, silicon carbide substrate **2200** comprises epitaxial layer **700** and epitaxial layer **1400** both formed of silicon carbide. FIG. **23** is not drawn to scale since thickness of remaining silicon carbide substrate **2230** is in the range of 300-400 micrometers, while the portion of silicon carbide substrate **2200** is in the range of 20-60 micrometers.

FIG. **24** is an illustration of silicon carbide substrate **2200** with Schottky Barrier Diode 1950 coupled to carrier wafer **1900** after removal by the exfoliation tool in accordance with an example embodiment. In one embodiment, silicon carbide substrate **2200** with Schottky Barrier Diode 1950 coupled to carrier wafer **1900** is processed to remove portions of vaporized or partially vaporized silicon carbide **2010** and vaporized or partially vaporized carbonized regions **2000** from FIG. **21**. Portions of vaporized or partially vaporized carbonized regions **2000** from FIG. **21** may be removed by etching in an oxygen plasma exposing portions of epitaxial layer **700**. In one embodiment, epitaxial layer **700** may be polished using chemical mechanical polishing after etching portions of vaporized or partially vaporized carbonized regions **2000** from FIG. **21** in an oxygen plasma.

In the example embodiment, a silicon carbide substrate of a predetermined thickness can be formed using the process disclosed herein above to improve thermal transfer and lower resistance of a silicon carbide device while lowering manufacturing cost.

FIG. **25** is an illustration of a metal layer **2500** deposited on a surface of epitaxial layer **700** in accordance with an example embodiment. In one embodiment, silicon carbide substrate **2200** is coated with metal layer **2500** to form a backside contact of Schottky Barrier Diode 1950. In one embodiment, the surface of epitaxial layer **700** of silicon carbide substrate **2200** is polished and metal layer **2500** is deposited on the surface of epitaxial layer **700** with good ohmic contact using evaporation, sputtering and other methods of metal deposition. Epitaxial layer **700** is formed with N+ doping to ensure good ohmic contact with metal layer **2500**. Metals such as nickel, or combination of metals such as Ti/Ni/Au (Titanium/Nickel/Gold) may be used along with annealing to reduce contact resistance to surface of epitaxial layer **700**. In one embodiment, laser annealing may be used to reduce contact resistance of metal layer **2500**.

FIG. **26** is an illustration of silicon carbide substrate **2200** with Schottky Barrier Diode 1950 separated from carrier wafer **1900** of FIG. **25** in accordance with an example embodiment. In one embodiment, after metal layer **2500** is deposited, the entire assembly comprising of silicon carbide substrate **2200** and carrier wafer **1900** is attached to a blue dicing tape. Carrier wafer **1900** is then separated from silicon carbide substrate **2200** which is coupled to the dicing tape. In the example embodiment, silicon carbide substrate **2200** with completed Schottky Barrier Diode 1950 is then diced and assembled in packages

FIG. **27** is an illustration of a remaining silicon carbide substrate **2230** separated from silicon carbide substrate **2200** from FIG. **26** after the exfoliation process in accordance with an example embodiment. Remaining silicon carbide substrate **2230** is separated from silicon carbide substrate

2200 with portions of vaporized or partially vaporized carbon region **2720** and vaporized or partially vaporized silicon carbide region **2710** on the surface of remaining silicon carbide substrate **2230**. Remaining silicon carbide substrate **2230** is a majority portion of silicon carbide substrate **100** from FIG. **1** after the exfoliation process and is further processed to make remaining silicon carbide substrate **2230** suitable for reuse.

FIG. **28** is an illustration of further processing of the remaining silicon carbide substrate **2230** separated from silicon carbide substrate **2200** from FIG. **26** after the exfoliation process in accordance with an example embodiment. Remaining silicon carbide substrate **2230** separated from silicon carbide substrate **2200** is polished with portions of vaporized or partially vaporized carbon region **2720** and vaporized or partially vaporized silicon carbide region **2710** removed from the surface of remaining silicon carbide substrate **2230** is further processed to make it suitable for reuse.

As previously disclosed herein above, silicon carbide substrate **2230** is a majority portion of silicon carbide substrate **100** from FIG. **20** and is reclaimed by re-polishing a surface exposed to fracture plane **2220** from FIG. **23** such that a polished surface is suitable for formation of semiconductor devices using the current invention. The polishing of the surface of silicon carbide substrate **2230** to form reclaimed silicon carbide substrate **2230** is performed using CMP (chemical mechanical polishing), electrochemical polishing among other methods. Reclaimed silicon carbide substrate **2230** can be used for successive formation of semiconductor devices using the same silicon carbide substrate **100** but with a portion removed by each subsequent exfoliation process.

By successive application of the current invention of formation of patterned carbon region **600** and plurality of silicon carbide pillars **710** from FIG. **7**, epitaxial growth of epitaxial layer **700**, epitaxial growth of drift region in epitaxial layer **1400**, device formation in epitaxial layer **1400**, exfoliation by using a laser beam to vaporize carbon region **600** and plurality of silicon carbide pillars **710** and re-polishing of the severed substrate, the reusable silicon carbide substrate **100** may be re-used multiple times. By the successive application of the current invention as described by the example embodiment, the same reusable silicon carbide substrate **100** can be used for fabrication of silicon carbide semiconductor devices leading to significant reduction in the cost of fabrication of silicon carbide semiconductor devices. By application of the exfoliation process using patterned carbon layer and laser exfoliation, silicon carbide devices can be fabricated with lower RDS (drain to source resistance) leading to higher electrical efficiency and lower thermal resistance.

FIG. **29** is an illustration of a block diagram **2992** of an exfoliation process **2990** in accordance with an example embodiment. Substrate forming process and exfoliation process **2990** supports reuse of semiconductor substrate in the manufacture of semiconductor devices. The order of the blocks in block diagram in FIG. **29** is for illustrative purposes only and does not imply an order or show all the specific steps in the implementation of the invention as are known by one skilled in the art.

In one embodiment, blocks **2900**, **2905**, **2910**, **2915**, **2920**, **2925**, **2930**, **2935**, **2940**, **2945**, **2950**, **2955**, **2960** and **2965** comprises the formation of a substrate and exfoliation process **2990** to separate the substrate from the reusable semiconductor substrate. In the example, the substrate comprises at least a first semiconductor epitaxial layer and a second semiconductor epitaxial layer and semiconductor devices are formed in the substrate. In one embodiment, no semiconductor devices are formed in the semiconductor substrate but the semiconductor substrate is used to form the substrate comprising at least two epitaxial semiconductor layers. In the block diagram **2992**, block **2900** illustrates the semiconductor substrate used in an example embodiment. In block **2905**, an array of pillars is formed in semiconductor substrate and gaps in pillars are filled with carbon as shown in block **2910**. After filling gaps in array of pillars with carbon, buffer epitaxial layer is formed as shown in block **2915** followed by forming of epitaxial drift layer, as shown in block **2920**. Buffer epitaxial layer and epitaxial drift layer comprise a semiconductor substrate. Block **2925** illustrates the step of forming at least one semiconductor device. Block **2930** shows the front side metallization of the at least one semiconductor device. Block **2935** shows the step of attaching the completed semiconductor device wafer with front side metallization to a UV transparent carrier wafer. The assembly of completed semiconductor device layer and carrier wafer is then subjected to the exfoliation process **2990**.

Block **2940** shows the exfoliation of substrate with at least one semiconductor device after exfoliation process **2990** using a laser to couple to patterned carbon layer in gaps of pillars formed in semiconductor substrate such that the semiconductor substrate is separated from the substrate comprising at least two semiconductor epitaxial layers.

Block **2945** shows semiconductor substrate comprising at least two semiconductor epitaxial layers with at least two semiconductor devices separated using an exfoliation tool. Block **2950** shows the step of polishing backside of the substrate followed by block **2955** showing the step of backside metallization of the substrate. Block **2960** shows the step of separating the substrate with completed semiconductor devices from the carrier wafer followed by block **2965** showing the step of testing and dicing of the substrate. In the example, a plurality of semiconductor devices is formed on or in the substrate and these are diced to separate the semiconductor devices for packaging. Block **2970** shows the portion of remaining semiconductor substrate after exfoliation of semiconductor device wafer as shown in block **2940**. Block **2975** of polishing remaining semiconductor substrate after exfoliation for reuse for multiple semiconductor devices and reuse as reusable semiconductor substrate for formation of semiconductor devices as shown in block **2980**. As mentioned herein above, only a fraction of the semiconductor substrate is used in the formation of the substrate. A remaining portion of the semiconductor substrate can be reused to form more substrates and more devices thus, extending the life of the semiconductor substrate and forming the devices on the substrate or a controlled and predetermined thickness.

FIG. **30** is an illustration of a block diagram **3092** of an exfoliation process **3090** in accordance with an example embodiment. Substrate forming process and exfoliation process **3090** supports reuse of semiconductor substrate in the manufacture of semiconductor devices. The order of the blocks in block diagram in FIG. **30** is for illustrative purposes only and does not imply an order or show all the specific steps in the implementation of the invention as are known by one skilled in the art.

In one embodiment, blocks **3000**, **3005**, **3010**, **3015**, **3020**, **3025**, **3030**, **3035**, **3040**, **3045**, **3050**, **3055**, **3060** and **3065** comprises the formation of a substrate and exfoliation process **3090** to separate the substrate from the reusable semiconductor substrate. In the example, the substrate comprises at least a first semiconductor epitaxial layer and a second semiconductor epitaxial layer and semiconductor devices are formed in the substrate. In one embodiment, no semiconductor devices are formed in the semiconductor substrate but the semiconductor substrate is used to form the substrate comprising at least two epitaxial semiconductor layers. In the block diagram **3092**, block **3000** illustrates the reusable silicon carbide substrate **100** used in an example embodiment. In block **3005**, an array of pillars **710** is formed in silicon carbide substrate and gaps in pillars are filled with carbon as shown in block **3010**. After filling gaps in array of pillars with carbon, buffer epitaxial layer **700** of silicon carbide is formed as shown in block **3015** followed by forming of epitaxial drift layer **1400** of silicon carbide, as shown in block **3020**. Buffer epitaxial layer and epitaxial drift layer comprise a semiconductor substrate. Block **3025** illustrates the step of forming at least one semiconductor device. Block **3030** shows the front side metallization **1800** of the at least one semiconductor device. Block **3035** shows the step of attaching the completed semiconductor device wafer in silicon carbide substrate with front side metallization to a UV transparent carrier wafer **1900**. The assembly of completed semiconductor device layer in silicon carbide substrate and carrier wafer is then subjected to the exfoliation process **3090**.

Block **3040** shows the exfoliation of silicon carbide substrate with at least one semiconductor device after exfoliation process **3090** using a laser **2030** to couple to patterned carbon layer in gaps of pillars formed in silicon carbide substrate such that the silicon carbide substrate is separated from the substrate comprising at least two semiconductor epitaxial layers.

Block **3045** shows silicon carbide substrate comprising at least two silicon carbide epitaxial layers with at least two semiconductor devices separated using an exfoliation tool. Block **3050** shows the step of polishing backside of the substrate followed by block **3055** showing the step of backside metallization **2500** of the substrate. Block **3060** shows the step of separating the substrate with completed semiconductor devices from the carrier wafer followed by block **3065** showing the step of testing and dicing of the substrate. In the example, a plurality of semiconductor devices is formed on or in the substrate and these are diced to separate the semiconductor devices for packaging. Block **3070** shows the portion of remaining silicon carbide substrate **2230** after exfoliation of semiconductor device wafer as shown in block **3040**. Block **3075** shows the step of polishing remaining silicon carbide substrate after exfoliation for reuse for multiple semiconductor devices and reuse as reusable silicon carbide substrate for formation of semiconductor devices as shown in block **3080**. As mentioned herein above, only a fraction of the silicon carbide substrate is used in the formation of the substrate. A remaining portion of the silicon carbide substrate can be reused to form more substrates and more devices thus, extending the life of the silicon carbide substrate and forming the devices on the substrate or a controlled and predetermined thickness.

While the present invention has been described with reference to certain preferred embodiments or methods, it is to be understood that the present invention is not limited to such specific embodiments or methods. Rather, it is the inventor's contention that the invention be understood and construed in its broadest meaning as reflected by the following claims. Thus, these claims are to be understood as incorporating not only the preferred methods described herein but all those other and further alterations and modifications as would be apparent to those of ordinary skilled in the art.

The descriptions disclosed herein below will call out components, materials, inputs, or outputs from FIGS. **1-30**.

In one embodiment, an exfoliation process comprises using a reusable silicon carbide substrate **100** with at least one silicon carbide epitaxial layer **700** and a patterned layer having a plurality of silicon carbide regions coupling first reusable silicon carbide substrate **100** to the at least one silicon carbide epitaxial layer **700** wherein patterned layer comprises a second material and wherein a laser **2030** is configured to heat the second material such that the plurality of silicon carbide regions in patterned layer **700** are vaporized or partially vaporized. In one embodiment, patterned layer comprises patterned carbon **600**. In one embodiment, the second material that forms a patterned layer comprises tantalum carbide.

In one embodiment, the exfoliation process wherein one or more devices are formed on or in the at least one silicon carbide epitaxial layer **700** and wherein the laser **2030** can couple through the at least one silicon carbide epitaxial layer **700** or the reusable silicon carbide substrate **100** to heat the second material of the patterned layer.

In one embodiment, the exfoliation process wherein a torque is applied to at least one of the reusable silicon carbide substrate **100** or the at least one silicon carbide epitaxial layer **700** to separate reusable silicon carbide substrate **100** from the at least one silicon carbide epitaxial layer **700** after the plurality of silicon carbide regions are vaporized or partially vaporized.

In one embodiment, the exfoliation process wherein a pulling force can be applied to the at least one of the reusable silicon carbide substrate **100** or the at least one silicon carbide epitaxial layer **700** to support the exfoliation process.

In one embodiment, the exfoliation process wherein the reusable silicon carbide substrate **100** is separated from the at least one silicon carbide epitaxial layer **700** and wherein reusable silicon carbide substrate **100** is prepared for reuse.

In one embodiment, the exfoliation process wherein each silicon carbide region of the plurality of silicon carbide regions in the patterned layer **600** is adjacent to the second material of the patterned layer.

In one embodiment, the exfoliation process wherein the second material comprises carbon.

In one embodiment, the exfoliation process wherein the at least one silicon carbide epitaxial layer **700** and the plurality of silicon carbide regions of the patterned layer are formed by merged epitaxial layer overgrowth on the reusable silicon carbide substrate **100** and wherein the at least one silicon carbide epitaxial layer **700** has a crystal orientation identical to the first reusable silicon carbide substrate.

In one embodiment, an exfoliation process for separating a reusable silicon carbide substrate **100** from at least one silicon carbide epitaxial layer **700** grown by merged epitaxial layer overgrowth (MELO) on reusable silicon carbide substrate **700** comprising a patterned layer between reusable silicon carbide substrate **100** and the at least one silicon carbide epitaxial layer **700** wherein a plurality of silicon carbide pillars **710** are formed in patterned layer when at least one silicon carbide epitaxial layer **700** is formed, wherein a laser **2030** is configured to heat a material in patterned layer such that the plurality of silicon carbide pillars **710** are vaporized or partially vaporized during the exfoliation process, and wherein a torque is applied to at least one silicon carbide epitaxial layer **700** or reusable silicon carbide substrate **100** to separate at least one silicon carbide epitaxial layer **700** from reusable silicon carbide substrate **100**.

In one embodiment, the exfoliation process wherein the laser **2030** has a wavelength in a range of 532 nanometers, 1064 nanometers, 623-700 nanometers, or 632 nanometers.

In one embodiment, the exfoliation process wherein the laser **2030** is pulsed or continuous wave.

In one embodiment, the exfoliation process wherein the material in the patterned layer is carbon.

In one embodiment, the exfoliation process wherein the laser **2030** is configured to heat the carbon in the patterned layer greater than 3000 degrees Celsius.

In one embodiment, the exfoliation process wherein the heat from the carbon is configured to drop by more than 3000 degrees Celsius within ten microns of the at least one silicon carbide epitaxial layer **700** and wherein at least one silicon carbide epitaxial layer **700** has a thickness greater than ten microns.

In one embodiment, the exfoliation process wherein a pulling force can be applied to the at least one of the reusable silicon carbide substrate **100** or the at least one silicon carbide epitaxial layer **700** to support the exfoliation process.

In one embodiment, a exfoliation process for separating substrates comprising a reusable silicon carbide substrate **100**, a patterned layer of carbon in or overlying a surface of reusable silicon carbide substrate **100**, at least one silicon carbide epitaxial layer **700** formed overlying patterned layer of carbon wherein a plurality of silicon carbide regions in patterned layer couple between reusable silicon carbide substrate **100** and at least one silicon carbide epitaxial layer **700** and wherein at least one epitaxial layer **700** has a crystal orientation of reusable silicon carbide substrate **100**.

In one embodiment, the exfoliation process wherein patterned layer **600** of carbon comprises a plurality of trenches **1100** configured to be formed in reusable silicon carbide substrate **100** and a plurality of microvoids **1110** configured to be formed underlying plurality of trenches **1100** wherein adjacent microvoids of plurality of microvoids **1110** do not couple together thereby forming the patterned layer, wherein plurality of microvoids **1110** are configured to be filled or to be partially filled with a polymer and pyrolyzed to carbon **1200**, and wherein silicon carbide between plurality of microvoids **1110** comprises the plurality of silicon carbide regions in patterned layer.

In one embodiment, the exfoliation process wherein the patterned layer of carbon comprises a layer of carbon **1200** deposited on the reusable silicon carbide substrate **100** and patterned by a photolithographic process wherein a merged epitaxial layer overgrowth process is configured to grow silicon carbide on exposed areas of the surface of the reusable silicon carbide substrate **100** through the patterned layer and form the at least one silicon carbide epitaxial layer **1300** and wherein the silicon carbide grown on the exposed areas of the surface of the reusable silicon carbide substrate **100** comprises the plurality of silicon carbide regions.

In one embodiment, the exfoliation process wherein a laser **2030** is configured to heat the carbon **1200** in the patterned layer such that the plurality of silicon carbide regions in the patterned layer is vaporized or partially vaporized.

In one embodiment, the exfoliation process wherein on or more devices **1950** are formed in or on the at least one silicon carbide epitaxial layer **700**, wherein a torque is applied to at least one silicon carbide epitaxial layer **700** or the reusable silicon carbide substrate **100** to separate at least one silicon carbide epitaxial layer **700** from reusable silicon carbide substrate **700**, wherein the patterned layer is configured to release under the shear force applied by the torque, and wherein the separated reusable silicon carbide substrate **2230** can be prepared and used in another exfoliation process.

In one embodiment, the exfoliation can be initiated using the laser prior to device fabrication with the Epitaxial layer attached to the substrate on the edges. Once the front side device fabrication is completed, the handle layer is attached and the edges can be released using a UV laser and the full exfoliation can be completed (please use the right language with the appropriate reference to the figures).

In one embodiment, an entire new thick substrate, from 100 microns to 400 Microns can be grown over the substrate with the release layer with carbon voids and then separated using the process described above. The approach to such growth can be Epitaxial growth following merged Epitaxial overgrowth or high temperature Chemical vapor deposition or physical vapor transport. (Please use the right language with the appropriate reference to the figures.

While the present invention has been described with reference to certain preferred embodiments or methods, it is to be understood that the present invention is not limited to such specific embodiments or methods. Rather, it is the inventor's contention that the invention be understood and construed in its broadest meaning as reflected by the following claims. Thus, these claims are to be understood as incorporating not only the preferred methods described herein but all those other and further alterations and modifications as would be apparent to those of ordinary skilled in the art.

What is claimed is:

1. An exfoliation process for separating substrates comprising:

providing a reusable silicon carbide substrate;

forming a patterned layer comprising a plurality of silicon carbide regions and a second material;

growing at least one silicon carbide epitaxial layer on or overlying the patterned layer;

heating the second material such that the plurality of silicon carbide regions are vaporized or partially vaporized; and

separating the at least one silicon carbide epitaxial layer from reuseable silicon carbide substrate.

2. The exfoliation process of claim **1** wherein one or more devices are formed in or overlying the at least one silicon carbide epitaxial layer.

3. The exfoliation process of claim **1** wherein a torque is applied to at least one of the reusable silicon carbide substrate or the at least one silicon carbide epitaxial layer to separate the reusable silicon carbide substrate from the at least one silicon carbide epitaxial layer after the plurality of silicon carbide regions are vaporized or partially vaporized.

4. The exfoliation process of claim **3** wherein a pulling force can be applied to the at least one of the reusable silicon carbide substrate or the at least one silicon carbide epitaxial layer to support the exfoliation process.

5. The exfoliation process of claim **4** wherein the reusable silicon carbide substrate is prepared for reuse.

6. The exfoliation process of claim **1** wherein each silicon carbide region of the plurality of silicon carbide regions in the patterned layer is adjacent to the second material of the patterned layer.

7. The exfoliation process of claim **1** wherein the patterned layer is formed by etching a plurality of trenches in the reuseable silicon carbide substrate and wherein the second material comprises carbon in the plurality of trenches.

8. The exfoliation process of claim **7** wherein the at least one silicon carbide epitaxial layer comprises a first silicon carbide epitaxial layer grown by epitaxial lateral overgrowth and one or more silicon carbide epitaxial layers grown by epitaxial vertical overgrowth, wherein the first silicon carbide epitaxial layer is grown on or overlying the patterned layer, and wherein the one or more silicon carbide epitaxial layers are grown on or overlying the first silicon carbide epitaxial layer.

9. An exfoliation process for separating substrates comprising:

providing a reuseable silicon carbide substrate;

etching a plurality of trenches in the reuseable silicon carbide substrate to form a patterned layer wherein the plurality of trenches forms a plurality of pillars;

depositing a material in the plurality of trenches of the patterned layer;

growing a first silicon carbide epitaxial layer on or overlying the patterned layer wherein the first silicon carbide epitaxial layer is grown by merged epitaxial lateral overgrowth (MELO);

forming a plurality of devices overlying the first epitaxial layer;

heating the material with one or more lasers wherein heat from the material vaporizes or partially vaporizes the plurality of pillars; and

separating the plurality of devices from the reuseable silicon carbide substrate.

10. The exfoliation process of claim **9** wherein the one or more lasers haves a wavelength in a range of 532 nanometers, 1064 nanometers, 623-700 nanometers, or 632 nanometers.

11. The exfoliation process of claim **10** wherein the one or more lasers is are pulsed or continuous wave.

12. The exfoliation process of claim **9** wherein the material comprises carbon or tantalum carbide.

13. The exfoliation process of claim **9** further including growing one or more silicon carbide epitaxial layers by epitaxial vertical overgrowth wherein the plurality of devices are formed in or overlying the one or more silicon carbide epitaxial layers.

14. The exfoliation process of claim **9** wherein exfoliation occurs along a plane of the patterned layer thereby leaving at least some of the patterned layer on the reuseable silicon carbide substrate and wherein the at least some of the patterned layer is removed from the reuseable silicon carbide substrate to prepare the reuseable silicon carbide substrate for reuse.

15. The exfoliation process of claim **9** further including coupling a carrier wafer to the plurality of devices wherein a pulling force or a torque can be applied to the reusable silicon carbide substrate or the carrier wafer to support separation of the plurality of devices from the reuseable silicon carbide substrate.

16. An exfoliation process for separating substrates comprising:

providing a reusable silicon carbide substrate;

forming a patterned layer in or overlying the reusable silicon carbide substrate wherein the patterned layer includes a plurality of silicon carbide regions and a material such as carbon or tantalum carbide;

growing a first epitaxial layer on or overlying the patterned layer wherein the first epitaxial layer is formed by epitaxial lateral overgrowth; and

growing one or more epitaxial layers on or overlying the first epitaxial layer wherein the one or more epitaxial layers are formed by epitaxial vertical overgrowth wherein one or more devices are formed in or overlying the one or more epitaxial layers; and

heating the material in the patterned layer with one or more lasers to weaken the patterned layer; and

separating the one or more devices from the reuseable silicon carbide substrate.

17. The exfoliation process of claim **16** further including coupling a carrier wafer to the one or more devices wherein a pulling force can be applied to the at least one of the reusable silicon carbide substrate or the at least one silicon carbide epitaxial layer to support the exfoliation process.

18. The exfoliation process of claim **16** wherein the carbon in the patterned layer was parylene or photoresist converted to carbon by pyrolysis.

19. The exfoliation process of claim **16** wherein the one or more epitaxial layers have the same crystal orientation as the reusable silicon carbide substrate and wherein the one or more epitaxial layers are grown from a surface of the first epitaxial layer comprising merged epitaxial lateral overgrowth.

20. The exfoliation process of claim **16** further including:

preparing a surface of the first epitaxial layer and the one or more epitaxial layers after separation for further wafer processing or packaging; and

preparing a surface of the reuseable silicon carbide substrate after separation for reuse wherein remnants of the patterned layer is removed.

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